

GPS Signal Modelling and Transmission Via Software Defined Radio

by

Haram Najam Khan
CUI/FA22-BCE-041/ATD
Laiba Qureshi
CUI/FA22-BCE-033/ATD

BS Thesis
In
Computer Engineering

COMSATS University Islamabad - Abbottabad Campus, Pakistan
Spring 2026

COMSATS University Islamabad - Abbottabad Campus

GPS Signal Modelling and Transmission Via Software Defined Radio

A Thesis Presented to
COMSATS University Islamabad - Abbottabad Campus

In partial fulfillment
of the requirement for the degree of

BS Computer Engineering

by
Haram Najam Khan
CUI/FA22-BCE-041/ATD
Laiba Qureshi
CUI/FA22-BCE-033/ATD

Spring 2026

GPS Signal Modelling and Transmission Via Software Defined Radio

An Undergraduate thesis submitted to the Computer Engineering Department as partial fulfillment of the requirement for the award of Degree of Bachelor of Science in Computer Engineering.

Name	Registration Number
Haram Najam Khan	CUI/FA22-BCE-041/ATD
Laiba Qureshi	CUI/FA22-BCE-033/ATD

Supervisor
Dr. Malik Summair Asghar
Co-Supervisor
Dr. Zahid Mahmood Jihangiri
Department of Computer Engineering
Abbottabad Campus
COMSATS University Islamabad (CUI) - Abbottabad Campus

August 2026

This thesis titled
GPS Signal Modelling and Transmission Via Software-Defined Radio

by
Haram Najam Khan (CUI/FA22-BCE-041/ATD)
Laiba Qureshi (CUI/FA22-BCE-033/ATD)

Has been approved
For the COMSATS University Islamabad, Abbottabad Campus

Supervisor: Dr. Malik Summair Asghar & Co-Supervisor: Dr. Zahid Mahmood Jihangiri
Department of Computer Engineering, CUI, Abbottabad Campus

HOD: Dr. Zahid Mahmood, Associate Professor
Department of Computer Engineering, CUI, Abbottabad Campus

Declaration

We, Haram Najam Khan (CUI/FA22-BCE-041/ATD), and Laiba Qureshi (CUI/FA22-BCE-033/ATD) hereby declare that we have produced the work presented in this thesis, during the scheduled period of study. We also declare that we have not taken any material from any source except referred to wherever due that amount of plagiarism is within an acceptable range. If a violation of HEC rules on research has occurred in this thesis, we shall be liable to punishable action under the plagiarism rules of the HEC.

Date: _____ Signature of the student:

_____

Haram Najam Khan
CUI/FA22-BCE-041/ATD
Laiba Qureshi
CUI/FA22-BCE-033/ATD

Certificate

It is certified that Haram Najam Khan (CUI/FA22-BCE-041/ATD), and Laiba Qureshi (CUI/FA22-BCE-033/ATD) has carried out all the work related to this thesis under my supervision at the Department of Computer Engineering, COMSATS University Islamabad, Abbottabad campus and the work fulfills the requirement for the award of BS degree.

Date: _____

_____

Supervisor:
Dr. Malik Summair Asghar
Co-Supervisor:
Dr. Zahid Mahmood Jihangiri
Department of Computer Engineering

_____

Head of Department:
Dr. Zahid Mahmood, Associate Professor
Department of Computer Engineering

Dedication

Firstly, we humbly dedicate this study to the Almighty Allah, expressing our heartfelt thanks for His constant guidance, strength, protection, and the skills bestowed upon us. We acknowledge with immense gratitude the gift of a healthy life and offer all these achievements to His divine grace.

This study is wholeheartedly dedicated to our parents and to our Supervisor and Co-Supervisor, Dr. Malik Summair Asghar and Dr. Zahid Mehmood Jihangiri, as well as the staff, for their unwavering support and invaluable guidance throughout our BS studies. Their continuous, helpful, and inspiring feedback has been instrumental in shaping our academic journey.

We extend our deepest gratitude and respect to our families, especially our parents and siblings, whose unwavering moral, spiritual, and financial support has been the cornerstone of our success. Their encouragement and sacrifices have paved the path to our achievements.

Acknowledgment

Our strongest gratitude goes to Dr. Summair Asghar, Dr. Zahid Jehangiri our distinguished Supervisor and Co-Supervisor for their valuable perspectives and consistent encouragement throughout this project, GPS signal modelling and transmission through SDR. Your professional knowledge, furnishing fundamental insights for our development in this research field.

Moreover, we have a dept of gratitude to our family, whose support and guidance were pivotal in the completion of this tenure within the defined time frame. Their unshakable dedication fueled us with determination and significantly contributed to our milestones.

I am also thankful to the faculty members of my department for providing the facilities and academic environment that made this work possible. My sincere appreciation extends to my classmates and friends, who offered their assistance, motivation, and constructive suggestions during various stages of this project.

We're proud to have completed this project, which demonstrates our commitment to knowledge advancement and passion for this field. We appreciate the efforts of all individuals involved, who contributed to the completion of this project successfully.

Haram Najam Khan **CUI/FA22-BCE-041/ATD**
Laiba Qureshi **CUI/FA22-BCE-033/ATD**

Abstract

This project is about creating a system that can make and send GPS L1 signals using a kind of radio called Software Defined Radio. Normally it is hard to work with GPS signals because they are very weak and hard to detect when you're inside a building. Also there are rules that say you cannot send out signals that're like GPS signals. So this project sets up a place in a laboratory where we can make and send GPS L1-like signals without causing any problems with the real GPS system that people use.

The system is designed to make signals that're very accurate and have good timing so they are like the signals that GPS satellites send out. This helps us get signals and analyse them properly. We use hardware and computer techniques to make sure the signals are sent out in a controlled way. When we did some tests the signals we made were very similar to the GPS signals that satellites send out. We can use this system to test GPS signals when we're inside a building and it is a good way to make sure receivers are working properly and to study signal processing. It is also good for people who are studying and doing research. This project shows that using Software Defined Radio to model GPS signals is a way to study how GPS signals work and to develop new receivers.

The GPS L1 signal generation and transmission system is very useful for this kind of work. The system is safe and controlled which is important when you are working with GPS signals. Overall the project is, about the GPS L1 signal generation and transmission system. How it can be used to study GPS signals.

TABLE OF CONTENTS

Title Page i
Inner Title Page i
Thesis Statement Page i
Thesis Approval Page ii
Declaration iii
Certificate iv
Dedication vi
Acknowledgment vii
Abstract vii

1 Introduction xvii
1.1 Motivation xix
1.1.1 Increasing dependence on GPS xx
1.1.2 Advantages of multi-band GPS xx
1.1.3 Need for a laboratory testbed xx
1.2 Problem Statement xxi
1.3 Objectives xxi
1.4 Significance of the Project xxii
1.5 Organization of the Thesis xxii

2 Literature Review xxiii
2.1 Software-Defined Radio in GNSS Research xxiii
2.2 SDR-Based GNSS Signal Generation xxiv
2.3 Multi-Band GPS Signal Processing xxv
2.4 RINEX-Based GNSS Data Processing xxv
2.5 GPS L1 Signal Modelling xxvi
2.6 GNSS Signal Recording, Replay, and Controlled Testing . . xxvii
2.7 GNSS Testbeds for Receiver Evaluation and Signal Security xxvii
2.8 Research Gap xxx
2.9 Critical Analysis of the Reviewed Literature xxx

3 Methodology and System Implementation xxxii
3.1 Development Strategy xxxii
3.2 Hardware and Software Environment xxxii
3.2.1 Hardware Environment xxxii
3.2.2 RF Cables and Attenuators xxxiv
3.2.3 Software Environment xxxiv
3.2.4 Overall Environment xxxv
3.3 RINEX File Parser VI xxxv
3.3.1 Developed RINEX File Parser VI xxxv
3.3.2 Processing Sequence xxxvii
3.3.3 Block Diagram of the RINEX File Parser VI xxxviii
3.3.4 Main Processing Stages xxxviii
3.3.5 Output of the RINEX File Parser VI xxxix

3.4 GPS Data Sorting VI xxxix
3.4.1 Working of GPS Data Sorting VI xxxix
3.4.2 Processing Sequence xl
3.4.3 Block Diagram Of GPS Data Sorting VI xli
3.4.4 Main Processing Stages xli
3.4.5 Output of the GPS Data Sorting VI xlii
3.5 PRN C/A Code Generator VI xlii
3.5.1 Background of PRN and C/A Code xliii
3.5.2 Developed PRN C/A Code Generator VI xliii
3.5.3 LabVIEW–Python Implementation xliv
3.5.4 Working of the PRN C/A Code Generator VI xliv
3.5.5 Python Backend for PRN Generation xlvii
3.5.6 Block Diagram of PRN C/A Code Generator VI . . . xlvii
3.5.7 Processing Sequence xlviii
3.5.8 Output of the PRN C/A Code Generator VI xlviii
3.6 Navigation Message Generator VI xlix
3.6.1 Navigation Message in the GPS L1 Signal xlix
3.6.2 TCP-Based Navigation Message Approach xlix
3.6.3 Developed Navigation Message Generator VI xlix
3.6.4 Working of the Navigation Message Generator VI . . l
3.6.5 TCP-Based Navigation Message Architecture lii
3.6.6 Navigation Data Rate lii
3.6.7 Block Diagram of the Navigation Message Generator VI lii
3.6.8 Main Functions of the Navigation Message Generator VI liii
3.6.9 Output of the Navigation Message Generator VI . . . liii
3.7 Carrier Signal Generation VI liii
3.7.1 Role of the Carrier in GPS L1 Signal Generation . . . liv
3.7.2 Carrier Parameters liv
3.7.3 Mathematical Representation of the Carrier liv
3.7.4 Developed Carrier Signal Generation VI lv
3.7.5 Working of the Carrier Signal Generation VI lv
3.7.6 Block Diagram of the Carrier Signal Generation VI . lvi
3.7.7 Processing Sequence lvi
3.7.8 Output of the Carrier Signal Generation VI lvii
3.7.9 Integration with the GPS L1 Signal-Generation System lvii
3.8 Final GPS Signal Integration Using NI GPS Data Streaming VI lvii
3.8.1 Need for the Integrated NI Implementation lviii
3.8.2 NI GPS Data Streaming VI lviii
3.8.3 Relationship Between the Developed VIs and the NI GPS Data Streaming VI lix
3.8.4 Overall Working of the NI GPS Data Streaming VI . lix
3.8.5 GPS Configuration Stage lix
3.8.6 USRP Transmission Configuration lx
3.8.7 GPS Data Generation Loop lxi
3.8.8 Producer–Consumer Architecture lxii
3.8.9 Purpose of the Producer–Consumer Architecture . . . lxii

3.8.10 Producer Loop Working lxii
3.8.11 GPS Data Queue lxiii
3.8.12 Consumer Loop Working lxiii
3.8.13 Waveform Monitoring lxiv
3.8.14 RF Signal Streaming lxiv
3.8.15 Generation Control and System Monitoring lxiv
3.8.16 Shutdown and Queue Destruction lxv
3.8.17 Comparison with the Developed Modular VIs lxvi
3.8.18 Modular Development and Integrated Streaming . . . lxvi
3.8.19 Complete Working Flow of the Final Integration . . . lxvi
3.8.20 Significance of the Integrated NI GPS VI lxvi
3.9 Final Hardware Integration and GPS L1 Transmission lxvii
3.10 Development and Verification Approach lxviii

4 Results, Verification, Challenges and Discussion lxx
4.1 Results of RINEX Navigation Data Processing lxx
4.1.1 RINEX File Parser Output lxx
4.1.2 Result Interpretation lxxi
4.2 Results of GPS Data Sorting VI lxxii
4.2.1 Sorted Satellite Navigation Data lxxii
4.2.2 Result Interpretation lxxiii
4.3 Results of PRN C/A Code Generator VI lxxiii
4.3.1 PRN Code Generation lxxiii
4.3.2 Python Backend and CSV Output lxxiv
4.3.3 Significance of the Output lxxiv
4.4 Results of Carrier Signal Generator VI lxxv
4.4.1 Carrier Waveform Generation lxxv
4.4.2 Verification Against LabVIEW Reference lxxv
4.5 Results of Navigation Message Conversion and TCP Streaming VI lxxvi
4.5.1 Navigation Word Reception lxxvi
4.5.2 Navigation-Bit Waveform lxxvii
4.5.3 Server-Side Result lxxviii
4.5.4 Result Interpretation lxxviii
4.6 Final GPS L1 Signal Integration Using NI GPS Data Streaming VI lxxviii
4.6.1 Integrated GPS Generation lxxviii
4.7 NI GPS Data Streaming VI Configuration Results lxxix
4.7.1 GPS Scenario Configuration lxxix
4.7.2 Interpretation of GPS Scenario Configuration lxxx
4.8 USRP Transmission Configuration lxxx
4.9 Producer–Consumer Architecture and Streaming Result . . . lxxx
4.9.1 Producer Loop lxxx
4.9.2 Consumer Loop lxxxi
4.9.3 Result Interpretation lxxxi
4.10 Final Generated GPS L1 Waveform lxxxi
4.10.1 Time-Domain Waveform lxxxi
4.10.2 Interpretation of the Waveform Shape lxxxii

4.11 Zoomed View of the Generated Waveform lxxxii
4.12 USRP-2920 Transmission Result lxxxii
4.13 Spectrum Analyzer Verification lxxxiii
4.14 Receiver-Level Validation lxxxiv
4.14.1 Satellite Detection lxxxiv
4.14.2 Position Fix Result lxxxiv
4.14.3 Significance of the 3D Fix lxxxv
4.15 Overall Experimental Verification lxxxvi
4.16 Challenges Encountered and Their Resolution lxxxvi
4.16.1 RINEX Data Handling lxxxvi
4.16.2 Navigation Message Processing lxxxvi
4.16.3 Integration of Individually Developed VIs lxxxvii
4.16.4 Continuous Data Streaming lxxxviii
4.17 Discussion of Experimental Results lxxxviii
4.18 Limitations of the Experimental Results lxxxix

5 Conclusion . . . . . **xc**

A Python Backend Implementation of the PRN C/A Code Generator . . . . . **xci**

LIST OF FIGURES

Figure 1.1 Basic concept of GPS-based positioning. xvii
Figure 1.2 GPS L1, L2, and L5 frequency bands. xviii

Figure 2.1 GPS multi-band frequency structure showing the L1, L2, and L5 bands. xxv
Figure 2.2 Conceptual role of RINEX navigation data in GNSS signal generation.. xxvi
Figure 2.3 Conceptual GPS L1 C/A signal-generation chain. xxvii
Figure 2.4 Conceptual architecture of SDR-based GNSS signal recording and replay. xxviii
Figure 2.5 Research landscape and positioning of the present GPS L1 signal-modelling work. xxxi

Figure 3.1 NI USRP-2920 software-defined radio xxxiii
Figure 3.2 NI USRP-2920 showing the major ports and interfaces. . . . xxxiii
Figure 3.3 GPS receiver output displayed using the GPSTest application on a mobile device. xxxiv
Figure 3.4 RF cables used for signal connections in the experimental setup xxxiv
Figure 3.5 Complete RINEX File Parser VI implemented in LabVIEW, shown in two parts. xxxvi
Figure 3.6 LabVIEW block diagram of the RINEX File Parser VI for extracting satellite navigation parameters from RINEX data. . xxxviii
Figure 3.7 Complete GPS Data Sorting VI implemented in LabVIEW. . xl
Figure 3.8 LabVIEW block diagram of the GPS Data Sorting VI. xli
Figure 3.9 Complete PRN C/A Code VI implemented in LabVIEW. . . xliv
Figure 3.10 Flowchart of the GPS C/A PRN code generation process. . . xlv
Figure 3.11 Block Diagram of PRN C/A Code Generation VI. xlviii
Figure 3.12 TCP Based Navigation Message Architecture for GPS L1 Signal Generation. l
Figure 3.13 TCP-based Navigation Message VI implemented in LabVIEW. l
Figure 3.14 Block Diagram of Navigation Message VI. lii
Figure 3.15 Carrier Signal Generation VI in LabView. lv
Figure 3.16 Block Diagram Of Carrier Signal Generation VI. lvi
Figure 3.17 Front panel of the NI GPS Data Streaming VI showing GPS and USRP configuration and generation controls. lix
Figure 3.18 GPS configuration section of the NI GPS Data Streaming VI. lxi
Figure 3.19 USRP transmission configuration section showing device, carrier frequency, IQ rate, gain, frequency offset, and reference frequency settings. lxi
Figure 3.20 Producer–Consumer architecture used for continuous GPS waveform generation and streaming. lxii
Figure 3.21 Producer loop responsible for generating GPS waveform blocks and placing them into the GPS data queue. lxiii

Figure 3.22 Consumer loop for retrieving generated GPS waveform blocks and transferring them toward the RF signal-generation interface. lxiv
Figure 3.23 Queue Destruction and Shutdown destroys the data queue, stops signal generation, updates the system state, and reports any errors. lxv
Figure 3.24 Overall experimental setup for transmission of the LabVIEW-generated GPS L1 signal through the USRP-2920 and transmit antenna to a mobile phone. lxvii

Figure 4.1 Output of the RINEX File Parsing VI showing the extracted satellite navigation parameters. lxxi
Figure 4.2 Output of the GPS Data Sorting VI showing the organized satellite navigation data. lxxii
Figure 4.3 Output of the GPS C/A PRN code generation VI showing the generated PRN sequence. lxxiii
Figure 4.4 CSV representation of the generated GPS C/A PRN codes produced by the Python implementation. lxxiv
Figure 4.5 Generated GPS L1 carrier signal waveform. lxxvi
Figure 4.6 Received GPS navigation word displayed as a U32 value in the Navigation Message VI. lxxvii
Figure 4.7 Navigation-bit waveform obtained after conversion of the received GPS word into individual binary bits. lxxvii
Figure 4.8 Server-side execution showing the generation and TCP streaming of GPS navigation words. lxxviii
Figure 4.9 Generated GPS signal waveform in the time domain. lxxxi
Figure 4.10 Zoomed View of Generated GPS signal waveform in the time domain. lxxxii
Figure 4.11 Spectrum analyzer measurement showing the transmitted GPS L1 signal. lxxxiii
Figure 4.12 Satellite detection at the GPS receiver level. lxxxiv
Figure 4.13 Satellite detection at the GPS receiver level. lxxxv
Figure 4.14 Three-level experimental validation of the developed GPS L1 signal generation and transmission system. lxxxix

Figure A.1 Python backend code used for the PRN C/A Code Generator VI. xci

LIST OF TABLES

Table 2.1 Comparison of Selected Recent GNSS Research Approaches . xxix

Table 3.1 Main processing stages of the RINEX File Parser VI. xxxix
Table 3.2 Main processing stages of the GPS Data Sorting VI. xlii
Table 3.3 Main functions of the developed PRN C/A Code Generator. . xlviii
Table 3.4 Main processing stages of the Navigation Message Generator VI. liii
Table 3.5 Main processing functions of the Carrier Signal Generation VI. lvii
Table 3.6 Relationship between the developed GPS processing modules and the integrated NI GPS Data Streaming VI. lx
Table 3.7 Comparison between modular GPS processing and integrated GPS streaming functions. lxvi

Table 4.1 Carrier Signal Generator module-level test parameters. lxxv
Table 4.2 Summary of experimental results and their significance. . . lxxxvii

LIST OF ABBREVIATIONS

AI	Artificial Intelligence
BPSK	Binary Phase Shift Keying
C/A	Coarse/Acquisition
CSV	Comma-Separated Values
GPS	Global Positioning System
G1	First Linear Feedback Shift Register used in GPS C/A code generation
G2	Second Linear Feedback Shift Register used in GPS C/A code generation
IQ	In-phase and Quadrature
L1	GPS L1 Frequency Band
L2	GPS L2 Frequency Band
L5	GPS L5 Frequency Band
LabVIEW	Laboratory Virtual Instrument Engineering Workbench
PRN	Pseudo-Random Noise
RF	Radio Frequency
SDR	Software Defined Radio
USRP	Universal Software Radio Peripheral
VI	Virtual Instrument

Chapter 1

Introduction

Today, positioning by satellite is an indispensable enabling technology for the digital infrastructure. Navigation and timing are currently used in many applications such as communication and timing, autonomous platforms, surveying, vehicles, aircraft and emergency-response technologies, all in the smartphones. GPS is one of main Global Navigation Satellite Systems (GNSS) to deliver Positioning, Navigation, and Timing (PNT) services, which allows GPS receivers to determine their location and to estimate accurate time based on signals sent by the satellites.

The GPS receiver is not just able to detect the one radio frequency sinusoidal carrier.



Figure 1.1: Basic concept of GPS-based positioning.

As illustrated in Figure 1.1, GPS positioning is based on signals transmitted by multiple satellites and received by a navigation receiver. In fact, the GPS signal received is a series of carefully structured components, such as the carrier signal, the satellite-specific spreading code, and navigation information. These components enable the receiver to recognize individual satellites, synchronize with the transmitted signal, obtain code and carrier timing, decode the navigation information and estimate the propagation delay between a satellite and the receiver. Various satellites can then be used to give measurements, which are then combined to provide position and timing information for the receiver.

The GPS signals are broadcast on several frequency bands, each with unique properties and benefits. The three frequency bands selected as part of this overall project include GPS L1 (1575.42 MHz); GPS L2 (1227.60 MHz); and GPS L5 (1176.45 MHz). The multiple frequency operation of the GPS is especially

valuable because the different frequencies are subject to different ionosphere effects. Several frequency measurements can thus be used to estimate and mitigate the ionospheric delay, enhance positioning accuracy and strengthen navigation solutions. L1 is the most popular GPS frequency and was the basis of the signal generation work presented in this thesis, whilst L2 and L5 are important extensions for future development of the proposed multi-band platform.

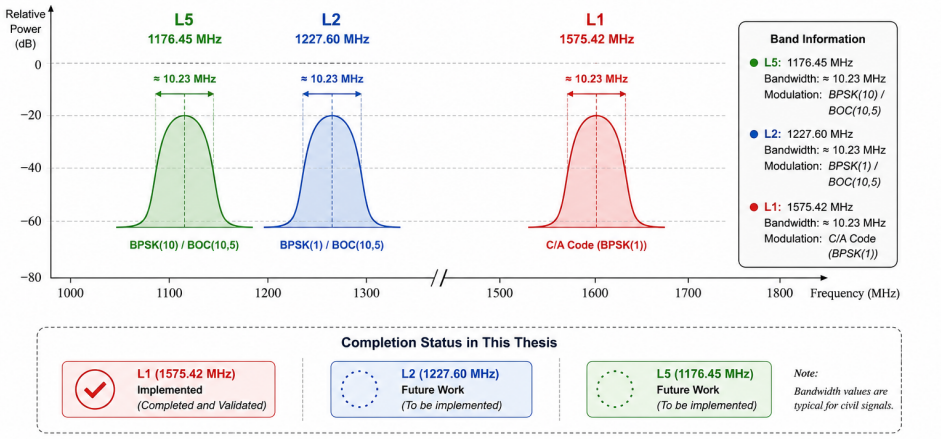


Figure 1.2: GPS L1, L2, and L5 frequency bands.

There are several practical difficulties to study GPS signals. Satellite signals at Earth's surface are very weak and can be hard to lock on to in indoor environments. Moreover, it is not easily possible to generate live GPS signals with free control of the satellite parameters, signal conditions and experiments. Multipath propagation, atmospheric propagation, satellite geometry, interference and environmental changes may also affect outdoor experiments. Furthermore, the unregulated transmission of GPS-like signals in the RF band being used for the operational services may hamper the actual navigation service, and is subject to regulations. The limitations of these provide a requirement for a controlled laboratory setting to reproduce the characteristics of a GPS signal in a safe and consistent way for academic and research applications.

One possible solution to these challenges is Software Defined Radio (SDR). Much of the signal processing is done digitally in an SDR system, rather than using dedicated hardware circuits. The signal components of GPS can then be generated, processed, modified and analyzed with software and then interfaced with SDR hardware for controlled RF experimentation. This is a method that allows to investigate the various phases of GPS signal generation without sacrificing the flexibility of a practical implementation with fixed function hardware.

The development of GPS signal-processing modules will use LabVIEW as the main software platform, and the SDR platform to be used for the intended RF signal transmission and analysis will be provided by the Universal Software Radio Peripheral (USRP). The whole system is structured modularly in the sense that each part of the GPS signal generation process is designed as an independent Virtual Instrument (VI). Each stage is independent and can be developed and tested in isolation before adding it to the overall signal chain.

The GPS L1 signal chain used in the current phase of the project consists of the extraction and organization of satellite navigation parameters from a RINEX

navigation file, the generation of the satellite specific Pseudo-Random Noise (PRN) C/A code and the generation of the carrier waveform, as well as the development of the Binary Phase Shift Keying (BPSK) modulation stage. The PRN code is used to identify the individual GPS satellites and the navigation data is used to decode what is being transmitted. These components are then added to the carrier to create the desired GPS-like waveform.

To facilitate implementation, the GPS signal generation algorithm is modularized, where each module is a processing step in the signal generation. First of all, the navigation data need to be acquired and structured, then the satellite-specific PRN code and carrier generated. The navigation data and spreading code are then mixed together and used to modulate the carrier with BPSK. This yields a digital waveform suitable for SDR transmission and analysis. These functions are easier to test, to see if things are done correctly, and to systematically combine the entire signal-processing chain as individual modules.

While the overall concept of the project is the development of a multi-band GPS platform using L1, L2 and L5 signals, this thesis concentrates on the GPS L1 signal. The L1 band was chosen as the main band for implementation because it maintains well-known signal structure and can be used as a base band for the development and validation of the basic generation techniques for the signals. The L2 band and L5 band are not part of the current project phase, but are future extensions of the proposed system. To be applicable to these bands, the present architecture would need to include the signal structures, spreading codes, navigation data characteristics, modulation schemes and associated processing requirements of each band.

The modular approach to development also offers a valuable educational and research benefit. A characteristic of each processing block is the ability to be seen and tested individually before integration, which enables study of the behaviour of the signal at various stages of the signal generation process. This makes the proposed platform not only as a demonstration platform of the generation of GPS signals but also an understanding platform of digital communication, spread-spectrum signalling, synchronization, and SDR-based waveform processing.

Hence, the main contribution of this work is the design of an analysis and modelling framework for GPS L1 signal generation using controlled SDR. The system put in place creates the basis for a multi-band GPS signal system and creates a helpful lab environment for experimentation with signal processing. The developed architecture can then be expanded to include L2 and L5, allowing for more comprehensive GPS experimentation in multiple frequencies, and future experimentation and testing of receivers, signal analysis, GPS positioning studies, and advanced GNSS signal processing techniques.

1.1 Motivation

With the growing use of Global Positioning System (GPS) technology in many modern navigation, communication, transportation, surveying, timing, and location-based applications, there is a great demand for reliable and well understood positioning infrastructure. The advent of GPS based systems in critical and everyday applications makes understanding the mechanisms of the GPS signal generation and processing important for both academic research and engineering

development. The practical limitations however are that the received GPS signals on the ground are very weak, the environment is constantly changing, and the experimental parameters are not directly controlled by the experimenter, when using live satellite signals. The challenges drive the creation of a controlled laboratory environment that allows GPS signal-processing concepts to be systematically studied.

1.1.1 Increasing dependence on GPS

GPS is now an integral part of many modern technologies such as vehicle navigation, aviation, surveying, telecommunication systems, infrastructure monitoring, timing and synchronization systems, autonomous platforms, and emergency response systems. Besides its importance for geographical location determination, accurate timing, which is also provided by GNSS systems, is important for many communication and electronic systems. So degradation, interruption, or misinterpretation of GPS signals can impact a variety of applications.

This ubiquitous reliance necessitates the development and understanding of the generation, transmission, acquisition and processing of GPS signals. The laboratory-based signal-generation platform offers an opportunity to study the processes under controlled conditions, not just using the unpredictable live satellite signal. It also serves as a basis for studying receiver behaviour, signal quality, signal synchronisation, and other signal processing concepts.

1.1.2 Advantages of multi-band GPS

Another important motivation for the project is the use of multiple GPS frequency bands. GPS signals are sent at various frequencies, such as 1575.42 MHz on L1, 1227.60 MHz on L2 and 1176.45 MHz on L5. The propagation of signals at various frequencies is affected differently by the propagation in the ionosphere. Thus, data from a number of frequencies can be used to supplement the information required for estimating and minimizing the ionospheric error, which enhances positioning performance and robustness.

Multi-band operation also offers the opportunity to study different signal shapes, modulation parameters, bandwidth requirements, acquisition methods and receiver processing techniques. The entire project was designed as multi-band GPS signal modelling and transmission platform for this reason. The implementation presented and tested in this thesis, however, only targets the L1 signal. L2 and L5 remain as future potential extensions of the architecture that has been developed.

1.1.3 Need for a laboratory testbed

This is because a programmable laboratory testbed offers many benefits compared to relying on live GPS reception. These satellite characteristics can not be controlled in an unconstricted manner with regard to the satellite to be used, the properties of the signal, the navigation data, the propagation environment, or the time to be used for experimentation. With a controlled signal generation environment, on the other hand, researchers can set and recreate specific experimental conditions.

A platform of this type can make it possible to choose a specific satellite PRN, to control the use of navigation data, to modify the parameters of the signals gen-

erated and to observe the intermediate steps of the process, and to analyse the generated waveforms repeatedly. This repeatability is especially beneficial in the debugging, validation, receiver testing, and academic experimentation.

The project is thus a motivation to develop a flexible and programmable base for signal modelling of GPS signals with LabVIEW and Software Defined Radio (SDR) methods. The project consists of GPS L1 signal-processing chain, which is implemented as modular software components, in a controlled environment where each stage of the processing chain, including navigation-data processing, PRN C/A code generation, carrier generation, and BPSK modulation, can be developed and analysed on its own. The final outcome can be used as a basis for further integration with SDR hardware and then expanded to an L2 and L5 platform, and used for further research and experimentation in GPS and GNSS signal processing.

1.2 Problem Statement

In spite of the fact that GPS is widely used in navigation and timing applications today, it is uncommon for university laboratories to have flexible and programmable systems for the generation of controlled GPS signals. The nature of live satellite signals and its associated parameters such as signal strength, satellite geometry, atmospheric conditions, multipath, and RF interference constantly vary and are difficult to recreate in experimental studies. Additionally, GPS signals are not easily tampered with to establish particular test environments and transmission of GPS-like signals can disrupt operational navigation services and is subject to regulation limitations.

Controlled signal generating capabilities can be provided by commercial GPS simulators, but they are costly and not readily available or adaptable to academic experimentation. This means a flexible modular framework and a laboratory scale model to be able to model and study the major stages of the GPS signal generation under controlled conditions.

The overall goal of this project is to create a multi-band GPS signal modelling platform for L1, L2 and L5 bands. The current implementation is to build and test the GPS L1 signal-processing chain, however. A key engineering challenge is to create a systematic approach for acquiring realistic satellite navigation parameters, organizing the parameters to process in LabVIEW, generating PRN C/A codes for satellite acquisition, building and modulating the GPS signal, and preparing the resulting signal for controlled SDR-based transmission and analysis.

1.3 Objectives

1. To learn about and analyze the structure and properties of GPS L1, L2, and L5 signals, including their carrier frequencies, PRN codes, navigation data, and modulation techniques.

2. To design and develop a GPS signal modelling system using modular LabVIEW-based processing blocks.

3. To obtain navigation data from RINEX files and extract the satellite parameters required for GPS signal generation.

4. To design the GPS L1 signal-processing chain, including PRN C/A code generation, carrier generation, navigation data processing, and BPSK modulation.

5. To implement the designed L1 signal-processing modules into a functional GPS L1 waveform generation system.

6. To prepare the generated L1 signal for controlled SDR-based transmission and analysis using the available USRP hardware.

7. To establish the developed L1 framework as a basis for future L2 and L5 implementation and further GPS signal analysis.

1.4 Significance of the Project

This project introduces the GPS signal generation, and processing chain in a practical way, instead of as a black-box positioning system. It builds a modular framework in LabVIEW for GPS L1 signal modelling, and lays the groundwork for control over signal transmission and analysis using SDR techniques – extendable to L2 and L5. The developed framework can also be used in future studies on GPS receiver testing, signal analysis, synchronization and controlled interference. The project also offers hands-on practice with digital signal processing (DSP), RINEX data handling, LabVIEW programming, generation of waveforms, modulation, debugging, and integration of the system.

1.5 Organization of the Thesis

1. In Chapter 1, the project is introduced, along with the motivation, problem statement, objectives, scope and significance of the project.

2. In Chapter 2, the theoretical framework and literature review are presented, which includes the GPS architecture, signal structure, L1/L2/L5 characteristics, PRN codes, navigation data, BPSK, SDR, synchronization, and signal integrity.

3. Chapter 3 describes the methodology and implementation for the LabVIEW modules created for the L1-oriented signal chain in detail.

4. The following Chapter 4 contains the results, verification observations, implementation status, challenges, and discussion on the current limitations of the modules.

5. The work is ended in Chapter 5, which summarizes the foundation achieved and sets out the future work, including the completion of L1 and extension to L2 and L5.

Chapter 2

Literature Review

The current development of the Global Navigation Satellite System (GNSS) has raised the demand for platforms that can be used to develop, evaluate, and validate navigation signals and receiver algorithms in a controlled environment. Traditional GNSS experimentation typically requires live satellite signals, thus restricting the ability of researchers to create identical experimental conditions or alter specific characteristics of the satellite signals. This has led to the emergence of software-defined radio (SDR), software-based signal simulation, hardware-in-the-loop (HIL) testing and multi-frequency GNSS platforms as research directions.

The research in the field of GNSS signal processing and experimentation that has occurred in recent years (2023-2025) has shown the following approaches. These are SDR based receiver development, software generated GNSS signals, RINEX based data processing, multi-band signal processing and controlled signal recording and replay. This chapter provides an overview of these developments, focusing on GPS L1, L2 and L5 signals, SDR-based architectures, RINEX navigation data, and laboratory-scale GNSS experimentation. The goal is to show what capabilities and limitations do current methods have and to describe the research gap that the current research fills.

2.1 Software-Defined Radio in GNSS Research

The software-defined radio is an important technology for GNSS research, as it enables signal-processing functions to be programmed and changed via software instead of fixed hardware. This flexibility enables researchers to test various signal structures, receiver algorithms, synchronization methods, and even new GNSS services, without having to design a completely new hardware architecture for each change.

Pany et al. [1] gave a detailed overview of the development of GNSS software-defined radio and explained the development history, existing implementations, and standardization efforts of GNSS SDR systems. The authors noted the need for programmable processing platforms in GNSS research, and described publicly available SDR implementations and intermediate-frequency sample-data standards. The review highlights that SDR has matured from an experimental idea to a well-developed research method for GNSS receiver development and signal processing.

In order to facilitate the development of experimental GNSS receivers in

the future, Majoral et al. [2] have designed a flexible System-on-Chip Field-Programmable Gate Array (SoC-FPGA) architecture. Their architecture features SDR flexibility, FPGA-based processing and is designed to enable multi-channel and multi-constellation GNSS experimentation. A platform was proposed and evaluated on static and low earth orbit scenarios, which showed good GNSS signal acquisition and tracking. The study is significant as it shows that programmable architectures can be applied as research testbeds instead of considering GNSS receivers as fixed architecture systems.

While Majoral et al. [2] concentrate on receiver development, the current project takes a different approach: SDR starts with the signal generation. The proposed system does not consider designing a full GNSS receiver, rather it focuses on the design of the digital GPS signal chain and the preparation of the resulting waveform for controlled SDR transmission.

2.2 SDR-Based GNSS Signal Generation

However, SDR platforms have been explored as well for the generation and transmission of controlled GNSS signals, in addition to receiver development. These are especially beneficial for systems where repeatable lab conditions are needed.

Forero et al. [3] proposed a hardware-in-the-loop (HIL) simulator for emulating the spacecraft orbital positioning system (OPS) with GNSS signals. They used their system to model the spacecraft trajectories and the corresponding GNSS signals in software, to add signal perturbations to the modeled signals, to modulate the modeled signals, and to broadcast the modulated signals using a software-defined radio to a commercial GPS receiver. The experimental results proved the capability of the receiver to determine the simulated position of the spacecraft. The study demonstrates that, based on SDR, it is possible to create a controlled GNSS testing environment with strong evidence.

The advantage of the approach presented by Forero et al. [3] is the software simulation of the RF hardware. The framework, however, is oriented towards spacecraft orbital-positioning experiments and HW-in-Loop evaluation. The architecture thus solves a larger simulation problem than the GPSSIMULATION problem addressed by the individual modules of GPSSIGNAL generation considered throughout the present work.

A more recent study by Ibrahem et al. [4] has studied the simulation and processing of GPS L1 signals for navigation in low-Earth-orbit satellites using SDR. They evaluated navigation performance in a controlled environment by relying on software-generated GPS L1 signals and SDR-based processing. The study shows the relevance of GPS L1 as a practical signal for SDR experimentation, and a direct example of software-generated signals of L1 being mixed with SDR technology.

The present project follows a similar overall approach to software generation of GPS L1 signals and transmission oriented towards SDR. The proposed implementation, however, is organized as a modular LabVIEW signal-processing chain, allowing development and evaluation of individual parts of the system (such as the satellite data processing, PRN generation, carrier generation, navigation data processing, and BPSK modulation).

2.3 Multi-Band GPS Signal Processing

Modern GPS operates at multiple frequencies, L1, L2 and L5, with different structures and thus multi-frequency measurements. In the last few years, therefore, multi-band processing has increasingly been taken into account, instead of only considering legacy L1 operation for GNSS receivers.

Kim et al. [5] created a multi-band signal processing-oriented GNSS receiver development platform. They were equipped with processing channels for GPS L1 C/A, L2C and L5 signals and signals from other constellations. Experimental testing revealed the successful processing of multiple GNSS signals and the capability of the proposed platform to develop and assess multi-band GNSS receiver algorithms.

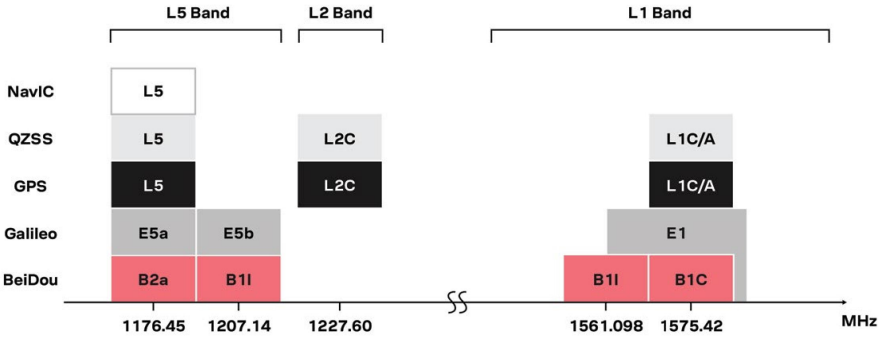


Figure 2.1: GPS multi-band frequency structure showing the L1, L2, and L5 bands.

In particular, the study of Kim et al. [5] illustrates the feasibility of integrating L1, L2, and L5 into a unified programmable processing system, a concept of great interest to the current project. The main goal of the multi-band receiver platform is, however, to have the development of the platform itself, while the current work focuses on the generation of the GPS signal itself. So, the overall direction of the present project is multi-band concept and only the L1 signal chain is applied.

Research on L5 has also highlighted the differences between modern L5 signals and the legacy L1 C/A signal. Kim et al. [6] studied the temporal blanking condition of the correlation properties of GNSS L5/E5a codes. They found that the correlation characteristics of L5/E5a codes are generally better than that of GPS L1 C/A codes, but this difference is reduced for large blanking duty cycles. The results show that the properties of each of the frequency bands has implications for signal acquisition, interference rejection, and reception processing.

The L5 research also lends support to the long-term interest of continuing the current implementation beyond L1. However, L2 and L5 are discussed as future development sections instead of implemented sections, due to the scope of the present project being the establishment and the validation of the L1 signal-processing chain.

2.4 RINEX-Based GNSS Data Processing

To generate realistic GNSS signals, the satellite information (i.e. ephemeris and clock parameters) needs to be correct. The Receiver Independent Exchange Format

(RINEX) is a standard way to store GNSS observation and navigation data and is widely used in GNSS research and processing applications.

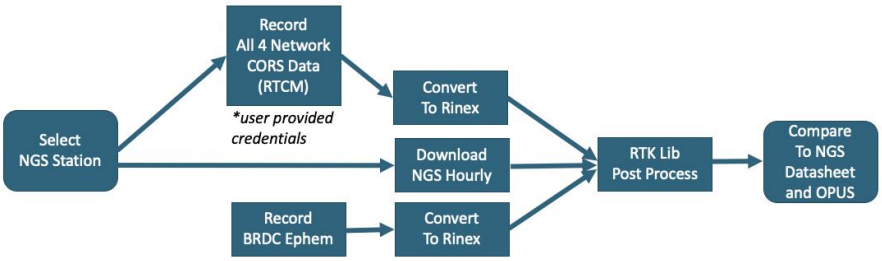


Figure 2.2: Conceptual role of RINEX navigation data in GNSS signal generation..

Kawamoto et al. [7] created RINGO, a RINEX preprocessing software package for handling multi-GNSS data. The system includes editing/merging, quality checking, clock-jump correcting, ionospheric correcting, and format conversion functions. The authors showed how structured RINEX preprocessing could be used to process ever more complex multi-GNSS data.

RINGO is more of a preprocessing tool than a GPS signal generator, but its efforts show that proper data handling of RINEX is crucial for GNSS applications. This is directly related to the present project since satellite navigation parameters are extracted from RINEX navigation data prior to its organization for subsequent GPS signal processing.

The ability to use RINEX data also has a benefit over the manual entry of satellite parameters. Navigation files are a well-documented representation of the satellite data which can be processed to yield the parameters needed for a signal-generation system. Consequently, the present project incorporates a RINEX data-processing stage before PRN generation, carrier generation, navigation-message construction, and modulation.

2.5 GPS L1 Signal Modelling

The L1 signal from GPS is one of the most extensive studied GNSS signal due to its well known signal structure and the broad receiver coverage. L1 C/A is the most practical of the available signals to be used for the development and testing of GNSS signal-generation systems.

Recent SDR research work still continues to use the GPS L1 signal as a representative signal for controlled experiments. Ibrahem et al. [4] have shown an SD approach to generate and process GPS L1 signals in an LEO navigation scenario. They have demonstrated how such an L1 signal can be generated by software and combined with SDR hardware to provide a versatile experiment platform.

The present project has a modular framework for L1 signal modelling. The signal generation process is broken down into separate functional blocks instead of being integrated as one component. The implemented and upcoming modules consist of RINEX data extraction, GPS data organization, PRN C/A code generation, carrier generation, navigation-data processing and BPSK modulation.

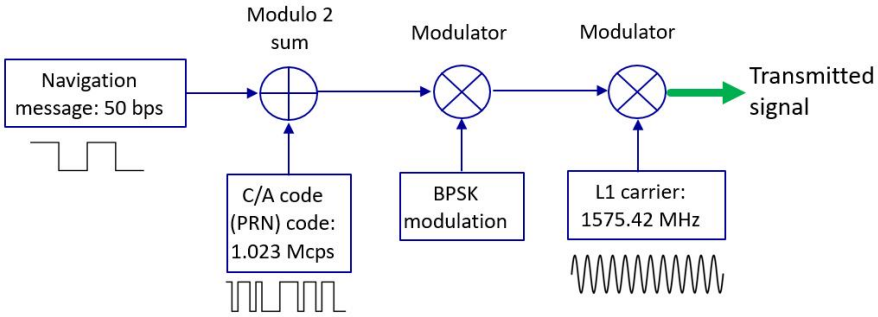


Figure 2.3: Conceptual GPS L1 C/A signal-generation chain.

It is modular, so that each processing step can be tested separately, which is useful for academic experimentation. This means that errors in PRN generation, carrier generation, data formatting, or modulation can thus be detected prior to the full integration of the signal. The method also lays the groundwork for future system expansion to other GPS frequency bands.

2.6 GNSS Signal Recording, Replay, and Controlled Testing

Another research direction is the development of controlled GNSS test environment by the recording and replay of GNSS signals. These methods enable researchers to simulate the actual signal conditions without having to depend on the availability of the same satellite geometry and environmental conditions for each experiment.

Stankevičius and Aleksiejūnas [8] created a low-cost SDR-based GNSS signal recorder/replayer for equipment testing. They implemented their system with GPS and GNSS L1 and L5 band receivers using time/phase synchronized HackRF One SDR modules. Multiple commercial GNSS receivers were used for experimental trials to show that the proposed system could recreate the recorded GNSS signals for experimental equipment testing. The authors also reported that, the experiment using a vehicle was conducted successfully in dynamic conditions.

This work is very applicable to the current project as it has shown that controlled multi-band GNSS experimentation is possible with relatively low cost SDR hardware. It is also a focus on timing and phase synchronization of GNSS signal reproduction.

But the record-and-replay system is quite different from a signal-modelling system. Record and replay involves capturing an existing RF environment, and then reproducing it. The present project is, on the other hand, to build up the GPS signal digitally using the satellite navigation parameters, PRN codes, navigation data and carrier generation stages. This allows more control of the individual signal components, and more suitable signal-generation study conditions.

2.7 GNSS Testbeds for Receiver Evaluation and Signal Security

Also, controlled GNSS environments are pivotal for the study of the receiver performance, interference and other GNSS environmental conditions that degrade the signal. These uses call for signal sources with controllable and repeatable characteristics.

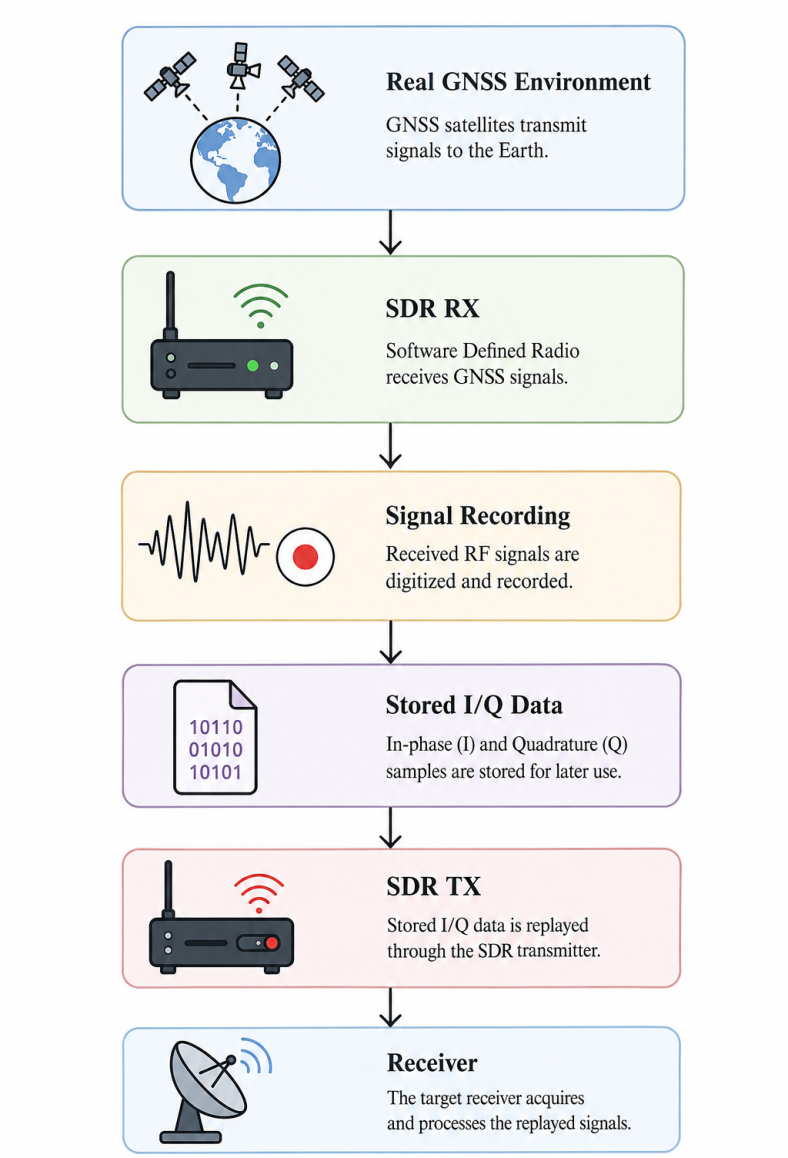


Figure 2.4: Conceptual architecture of SDR-based GNSS signal recording and replay.

Forero et al. [3] showed a platform of HW-in-the-loop, where they used a generated GNSS signal to transmit through SDR hardware to a real receiver. Such systems can be useful in the evaluation of receivers and development of algorithms, since a controlled signal scenario can be reproduced.

Likewise, research on signal replay of SDR signals has shown the effectiveness of such approach for testing commercial receivers under repeatable conditions [8]. The developments are a good sign that SDR platforms may become tools not only for communicating, but also for analyzing the GNSS signals and validating the receiver.

Table 2.1: Comparison of Selected Recent GNSS Research Approaches

Study	**Research Approach**	**Platform**	**Signal Focus**	**Main Contribution**
Forero et al. [3]	Hardware-in-the-loop simulation	SDR + GPS receiver	GNSS signals	Generated GNSS signals in software and transmitted them through SDR hardware to a commercial GPS receiver for spacecraft positioning experiments.
Majoral et al. [2]	Receiver development	SoC-FPGA	Multi-GNSS	Developed a flexible programmable architecture for the development and evaluation of experimental GNSS receivers.
Pany et al. [1]	Literature and technology review	GNSS SDR	Multi-GNSS	Reviewed the evolution of GNSS software-defined radio systems, sample-data standards, front-end technologies, and standardization efforts.
Kim et al. [5]	Multi-band receiver development	Programmable GNSS platform	L1, L2C, L5	Developed and tested a programmable platform supporting multiple GNSS frequency bands.
Kawamoto et al. [7]	RINEX data processing	Software	Multi-GNSS	Developed RINEX preprocessing capabilities for editing, checking, correcting, and managing GNSS data.
Ibrahem et al. [4]	Signal simulation	SDR	GPS L1	Investigated software-defined simulation and processing of GPS L1 signals for controlled GNSS experimentation.
Stankevičius and Aleksiejūnas [8]	Record and replay	HackRF One SDR	L1 and L5	Developed a low-cost SDR-based GNSS signal recording and replay system for controlled receiver testing.
Present Work	**Signal modelling**	**LabVIEW + USRP**	**GPS L1**	**Develops a modular RINEX-based GPS L1 signal-generation framework that can be extended to L2 and L5 in future work.**

The comparison indicates that other researchers have been successfully able to show some parts of the problem dealt with in this thesis. SDR development and signal transmission have been achieved with SDR platforms, structured GNSS data processing has been achieved with RINEX data and multi-band platforms have shown L1/L2/L5 signal processing. But these methods take different aims and architectures of implementation.

2.8 Research Gap

The literature reviewed confirmed that SDR is an effective technology for GNSS experimentation and recent research has shown software-defined receiver architectures, hardware-in-the-loop GNSS simulation, multi-band processing, RINEX preprocessing, and low cost recording/replay of signals. But these existing studies typically cover receiver development, full simulation frameworks, multi-band receiver platforms or reproduction of previously recorded signals.

There is a practical need for a modular, laboratory-based signal-generation structure with the individual components of the GPS signal construction being implemented, verified and validated independently. This type of architecture is particularly useful in an academic setting since the relationship between the satellite data, PRN codes, navigation information, carrier generation, and modulation can be studied in isolation.

This project aims to fill this gap, implementing a development of a LabVIEW based GPS L1 signal-modelling chain from the satellite navigation information provided by RINEX data. The architecture proposed divides the original signals generation tasks into independent modules that process the RINEX data, organize the GPS data, generate the PRN C/A code, generate the carrier, process the navigation data, and modulate the carrier by BPSK modulation. The output is a L1 waveform meant to be controlled and transmitted/analyzed with the available USRP SDR hardware.

Compared to multi-band studies which adopt full L1/L2/L5 receiver architectures, the focus of the current work is to first set up the implementation of the L1 signal chain. L2/L5 is kept for future additions to the developed framework. This step-by-step process decreases the complexity of implementation and helps to create a solid software basis for future multi-band development.

2.9 Critical Analysis of the Reviewed Literature

The literature shows that none of the approaches have all of the characteristics required for the present project. While software-only simulators may be highly flexible, they need not be directly related to controlled RF transmission. SDR receiver platforms offer versatility in programming and are mainly focused on the acquisition and tracking of a signal, as well as on navigation rather than the educational design of the transmitted signal itself. HW in the loop systems give realistic experimental environments but may consist of somewhat complex architectures. Record and replay systems can produce highly repeatable testing conditions, but instead of building the signal from the underlying navigation parameters, they reproduce previously captured signals.

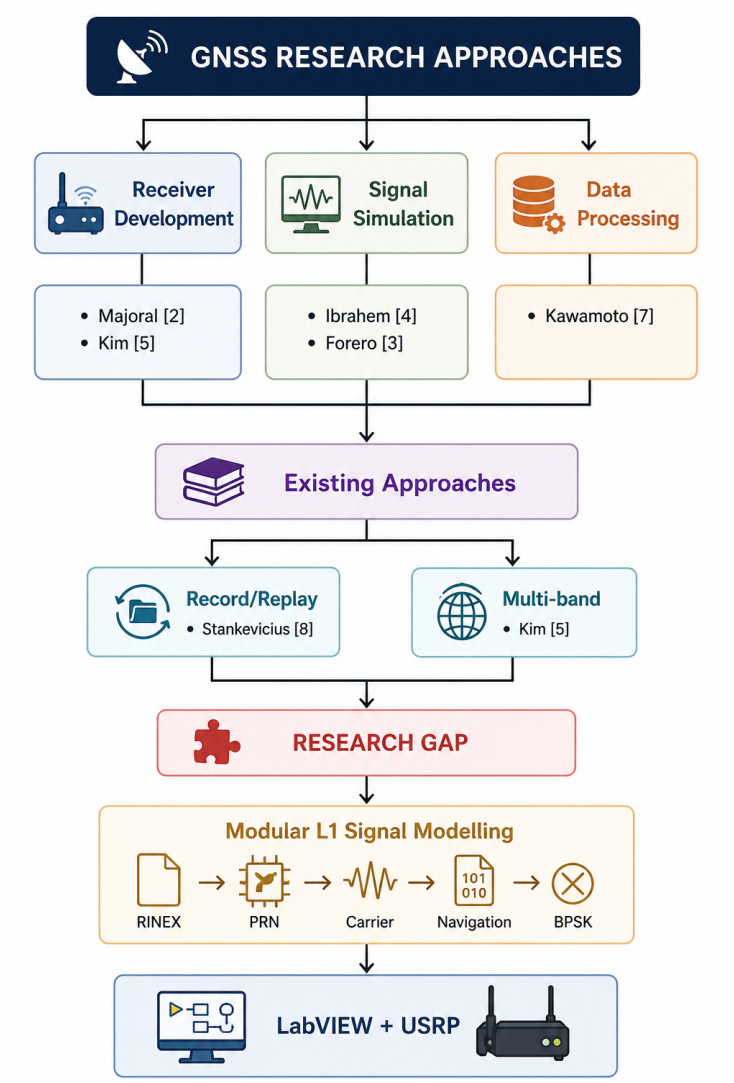


Figure 2.5: Research landscape and positioning of the present GPS L1 signal-modelling work.

The present work thus capitalizes on selected features of these approaches. The use of RINEX data offers some realistic satellite information, LabVIEW programming in a modular fashion gives insight into the individual stages of signal generation and SDR hardware gives a way from digital model of the waveforms to controlled RF experiments. The L1-focused implementation also offers a manageable framework to validate the fundamental signal generation process before moving up to L2 and L5.

Chapter 3

Methodology and System Implementation

3.1 Development Strategy

The modular engineering method was used to create the project. An entire processing chain was broken down into smaller VIs, rather than using one big GPS signal-generation VI. Each VI performs a specific function and the output of each VI is easily checked before being passed to the next processing VI. This makes debugging much easier and enables the intermediate result to be checked.

The execution order is dependent on the data dependencies of GPS signal. First, the information on the satellites is obtained from the RINEX navigation file. The information is then transformed into a structured format. Meanwhile the PRN C/A code can be generated for the corresponding satellite. The navigation data and PRN data are to be combined and transformed in a way to be able to interpret them in the further BPSK stage. The carrier waveform will be produced separately and utilized in the process of modulation.

3.2 Hardware and Software Environment

The final project configuration will be comprised of the hardware and software elements necessary to generate, process, transmit and validate the GPS L1 signal. The experimental setup features NI LabVIEW signal processing along with a software-defined radio (USRP-2920) and a GPS receiver, for laboratory validation. The major components are described separately, namely hardware and software, which is to distinguish the physical equipment from the signal-processing and control environment.

3.2.1 Hardware Environment

The hardware environment includes the software-defined radio, GPS receiver, RF connections and signal level management that were employed in the experimental setup.

USRP-2920 SDR

In the experimental setup, the last software-defined radio transmitter is the USRP-2920. It streams the generated GPS L1 waveform to the LabVIEW environment and provides the RF transmission interface for controlled lab validation.



Figure 3.1: NI USRP-2920 software-defined radio

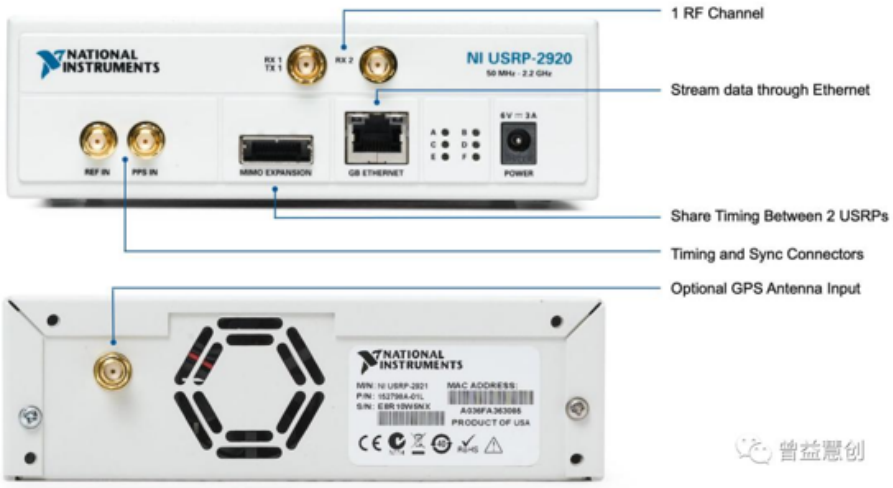


Figure 3.2: NI USRP-2920 showing the major ports and interfaces.

GPS Receiver

Final validation of the GPS L1 signal is done using a mobile GPS receiver. The receiver(s) are employed to detect the reception of the generated signal and to de-

termine the positioning information which is expected during the laboratory test.

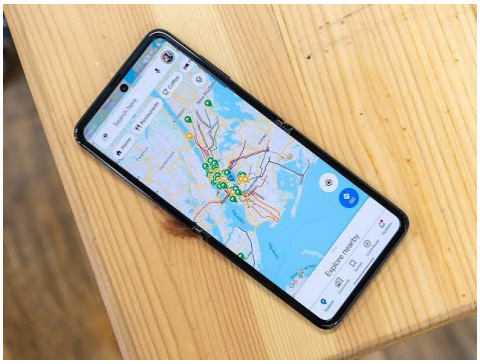


Figure 3.3: GPS receiver output displayed using the GPSTest application on a mobile device.

3.2.2 RF Cables and Attenuators

SMA coaxial cables with attenuators are used to connect the SDR to the receiving equipment.The SDR to receiver RF interconnection is provided by SMA cables and attenuators. The attenuators enable a controlled level management of the signal during lab testing purposes and an adequate RF input level for the receiver.



Figure 3.4: RF cables used for signal connections in the experimental setup.

3.2.3 Software Environment

The software environment offers the tools that are needed to process GPS data, generate the signal, modulate the signal and communicate with the SDR hardware.

NI LabVIEW

The GPS signal-processing VIs are written using the graphical programming environment, NI LabVIEW, as the main programming environment. The project is executed according to a modular VI based approach that allows to develop and test each processing stage separately.

NI-USRP Toolkit

The NI-USRP toolkit is a software interface between the LabVIEW environment and the USRP-2920. It is used to setup and control the SDR transmission stage as well as to push the created waveform to the USRP hardware.

3.2.4 Overall Environment

The hardware and software bring together to create a complete laboratory setup for signal generation, transmission and validation of the GPS L1 signal. The graphical programming environment for implementing and running the GPS signal processing VIs is provided by NI LabVIEW. NI LabVIEW provides the interface between the LabVIEW software and the NI-USRP software-defined radio, which is provided by the NI-USRP toolkit. The USRP-2920 is the experimental transmitter, and the RF cables and attenuators are used to provide controlled interconnection of signals. The transmitted GPS L1 signal's acquisition and positioning performance is validated by a GPS receiver.

3.3 RINEX File Parser VI

The first large data-processing module developed for the GPS L1 signal generation system is the RINEX File Parser VI. It is primarily used to read the RINEX navigation file and to extract the navigation parameters for the following stages of processing the GPS signal. The satellite-specific information contained in the RINEX file is needed for further processing of the navigation message, such as the satellite ephemeris and satellite clock parameters.

The RINEX parser is crucial because at the beginning the navigation information is provided in a **text-based file format** while the rest of the LabVIEW modules need the information in a structured numerical format. The VI serves as an external RINEX input/output and also as an intermediate link between the external RINEX navigation data and the internal GPS signal- generation system. Once the information is extracted, it is passed to the **GPS Data Sorting VI** where it is sorted into a suitable format for further processing.

3.3.1 Developed RINEX File Parser VI

The navigation file is selected and read, and the required navigation parameters are extracted and structured into a satellite navigation data structure using the developed **RINEX File Parser VI**.

Working

The **RINEX File Parser VI** is developed to read and process RINEX navigation files in LabVIEW. The VI is started by the user when the navigation file (required RINEX file) is chosen in the **File Path Control**. The file containing the selected file is then read into LabVIEW using the Read Delimited Spreadsheet VI. The RINEX file is a text file, so it can be analyzed to determine the various parts of the file and to retrieve the desired navigation data.

A RINEX navigation file contains a header and satellite navigation records. Thus, the entire file can not be used directly for satellite navigation. The VI looks for the word 'END OF HEADER' to separate these sections. This indicates the end of the header data and the start of the satellite navigation data. Upon identification of the marker, the header information is removed from the navigation data and only the satellite records relevant to the marker are processed.

The VI then sequentially processes the satellite records after it has located the navigation-data section. The **Scan From String** function is used to read the string and get the information from each record. The first line in each satellite record is the satellite's PRN number and **reference epoch** (year, month, day, hour, minute, second). The remaining lines of each satellite record contain orbital, clock, and other satellite-related parameters. These lines are processed sequentially using a **For Loop**. During each iteration, the required text values are extracted, converted into numerical values, and passed to the subsequent GPS processing stages.

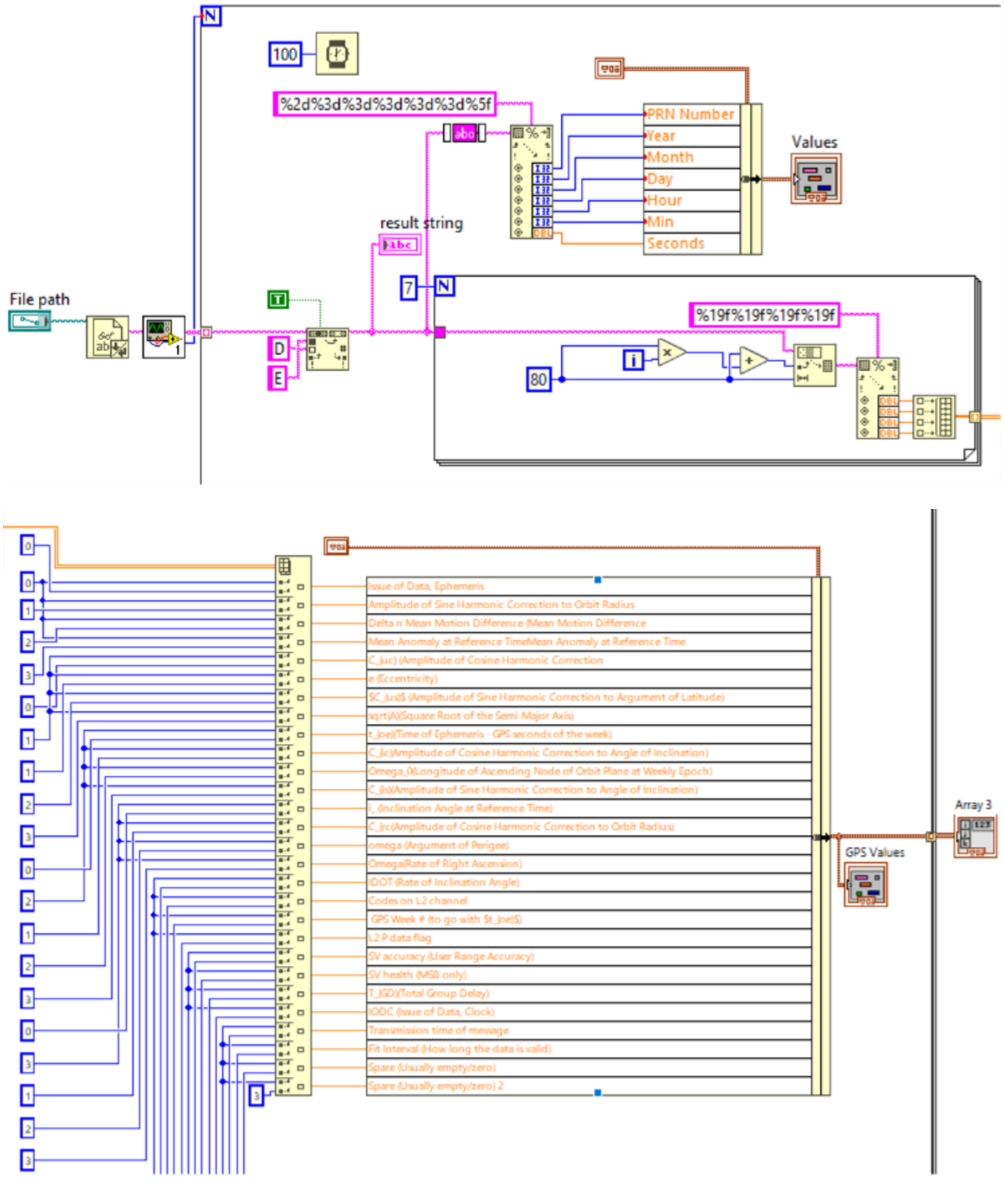


Figure 3.5: Complete RINEX File Parser VI implemented in LabVIEW, shown in two parts.

The main navigation parameters extracted by the VI include:

- Eccentricity
- Semi-major axis
- Inclination angle
- Longitude of the ascending node
- Argument of perigee
- Mean anomaly
- Harmonic correction parameters
- Satellite clock parameters
- GPS week and reference time parameters
- Satellite health and operational status information

The extracted parameters are converted into numerical values and stored in structured arrays and clusters within LabVIEW. This provides a consistent representation of the satellite navigation information and simplifies access to the required parameters.

The structured output of the **RINEX File Parser VI** is then provided to the next processing stage, the **GPS Data Sorting VI**. This allows the required satellite parameters to be accessed and organized efficiently for further GPS signal generation and processing.

3.3.2 Processing Sequence

A summary of the complete processing sequence of the developed RINEX File Parser VI is given below:

1. The required navigation file is selected by the file path control in the RINEX File Selection pop-up.
2. Read File Contents: The contents of the selected RINEX file are read into LabVIEW for processing.
3. The ”END OF HEADER” is identified and the satellite navigation records are separated from the header.
4. Separate Navigation Records: Records from the satellite navigation are extracted from the file.
5. Process Satellite Records: Satellite records are processed one after another.
6. Extract Required Navigation Parameters: String-processing functions are used to scan for the required navigation parameters.

7. Convert Text to Numerical Values: Extracted text values are converted into numerical parameter values.

8. The processed parameters are organized into structured arrays and clusters.

9. The structured GPS navigation data are sent to the GPS Data Sorting VI for further processing.

3.3.3 Block Diagram of the RINEX File Parser VI

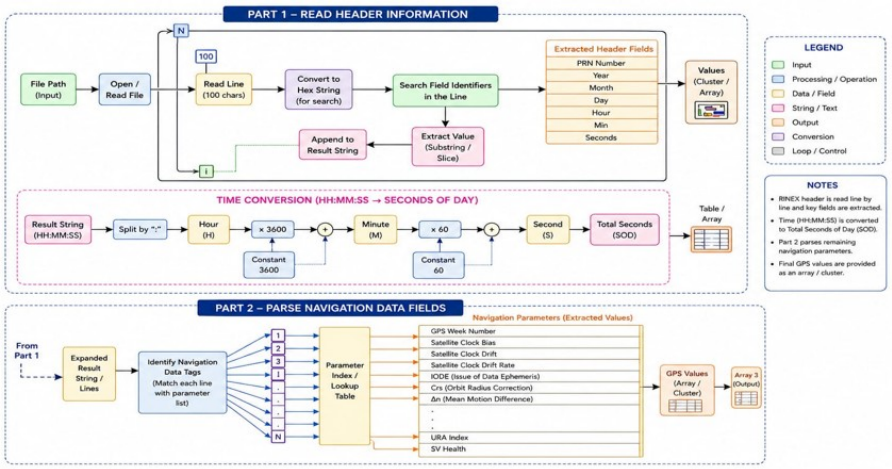


Figure 3.6: LabVIEW block diagram of the RINEX File Parser VI for extracting satellite navigation parameters from RINEX data.

3.3.4 Main Processing Stages

The main processing stages implemented in the RINEX File Parser VI are summarized in Table 3.1.

Table 3.1: Main processing stages of the RINEX File Parser VI.

Stage	Operation	Purpose
1	RINEX file selection	Selects the navigation file to be processed.
2	File reading	Loads the RINEX file contents into LabVIEW.
3	Header identification	Locates the "END OF HEADER" marker.
4	Navigation-data extraction	Separates satellite records from the header.
5	Record processing	Processes individual satellite records sequentially.
6	Parameter scanning	Extracts required values from the text records.
7	Numerical conversion	Converts extracted text into numerical parameters.
8	Data organization	Arranges the parameters into structured outputs.
9	Data transfer	Provides parsed data to the GPS Data Sorting VI.

3.3.5 Output of the RINEX File Parser VI

The output of the RINEX File Parser VI consists of the **extracted satellite navigation parameters in a structured form**. These outputs are not discussed in detail at this stage because their numerical values and resulting data structures are presented and analyzed in the **Results and Discussion** chapter.

The parsed information is subsequently supplied to the **GPS Data Sorting VI** for further organization and processing. This follows the project's overall processing sequence in which RINEX parsing is followed by GPS data sorting.

3.4 GPS Data Sorting VI

The **GPS Data Sorting VI** is a part of the system that generates GPS L1 signals. It does its job after the RINEX File Parser VI. The main job of the **GPS Data Sorting VI** is to get the satellite navigation information ready for the next step. This means it has to **organize and assign the parsed satellite parameters to their GPS ephemeris fields**.

The RINEX File Parser gets the information we need from the navigation file. At first, this information is just a list of numbers. We need to put these numbers into a format that makes sense. The **GPS Data Sorting VI** does this. It takes the list of numbers and puts them into a format that the next step can use.

3.4.1 Working of GPS Data Sorting VI

The **GPS Data Sorting VI** starts by looking at the navigation information it got from the RINEX processing stage. This information has the satellite navigation

records that we need to use. First, we have to separate these records from the rest of the information.

First, the **GPS Data Sorting VI** finds the **"END OF HEADER"** marker in the navigation data. This helps us separate the header from the satellite navigation records. Then we find out how many records there are.

Next, the **GPS Data Sorting VI** uses a **For Loop** to look at each record one by one. When it looks at a record, it breaks the record into pieces using **LabVIEW string-manipulation functions**. These pieces are like the parts of the satellite navigation record.

After that, the **GPS Data Sorting VI** puts the pieces into the right **GPS ephemeris fields**. This means that the right information, like the orbit and time, is in the place.

Finally, the **GPS Data Sorting VI** stores the sorted records in arrays. This makes it easy for the next step to use the information.

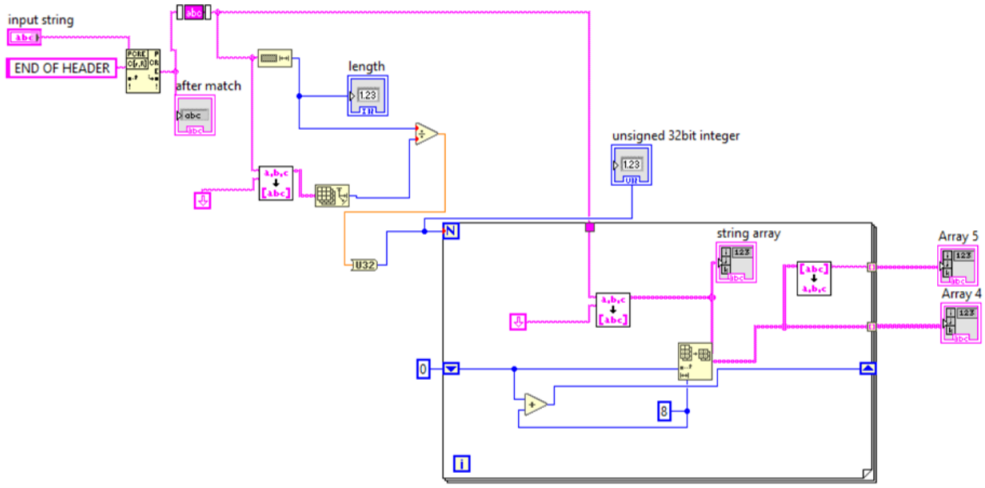


Figure 3.7: Complete GPS Data Sorting VI implemented in LabVIEW.

3.4.2 Processing Sequence

Here is how the **GPS Data Sorting VI** works:

1. **Get the navigation information:** The **GPS Data Sorting VI** gets the navigation information from the RINEX processing stage.

2. **Find the navigation records:** It separates the navigation records from the header information.

3. **Count the records:** It finds out how many navigation records there are.

4. **Look at each record:** The **GPS Data Sorting VI** uses a **For Loop** to look at each record one by one.

5. **Break the record into pieces:** It breaks each record into pieces using **LabVIEW string-manipulation functions**.

6. **Put the pieces in the place:** It puts the pieces into the right **GPS ephemeris fields**.

7. **Store the sorted records:** It stores the records in arrays.

8. **Send the information to the next step:** The **GPS Data Sorting VI** sends the navigation information to the next step.

3.4.3 Block Diagram Of GPS Data Sorting VI

The block diagram of the **GPS Data Sorting VI** shows how it works inside. It gets the navigation information, looks at each record, breaks the records into pieces, and puts the pieces into the place. This makes it easy for the next step to use the information. The **GPS Data Sorting VI** is a part of the system that generates GPS L1 signals.

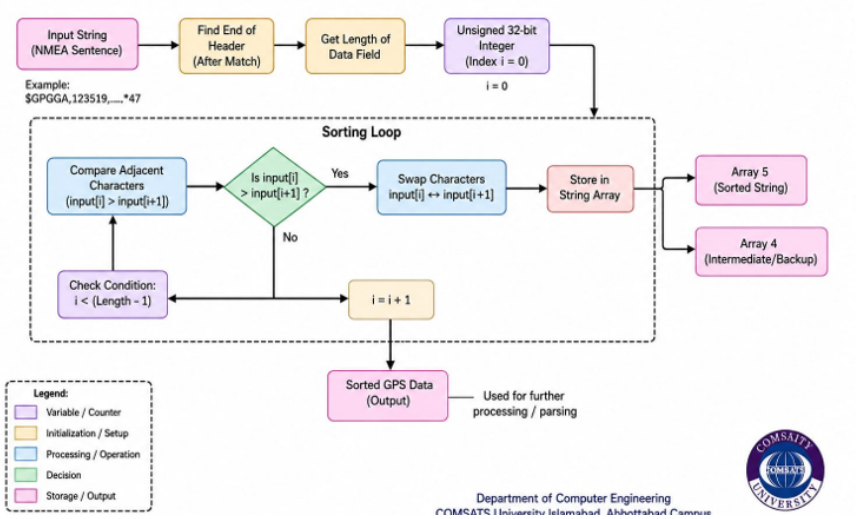


Figure 3.8: LabVIEW block diagram of the GPS Data Sorting VI.

3.4.4 Main Processing Stages

The main processing stages implemented in the GPS Data Sorting VI are summarized in Table 3.2.

Table 3.2: Main processing stages of the GPS Data Sorting VI.

Stage	Operation	Purpose
1	Receive parsed data	Accepts navigation information from the RINEX processing stage.
2	Navigation-section identification	Separates usable navigation records from the header information.
3	Record processing	Processes satellite records sequentially using a For Loop.
4	String separation	Divides each satellite record into smaller sections.
5	Parameter organization	Assigns extracted values to their corresponding GPS fields.
6	Array formation	Stores the organized satellite information in structured arrays.
7	Data output	Provides structured data to the navigation-message processing stage.

3.4.5 Output of the GPS Data Sorting VI

The output of this VI consists of **organized satellite navigation data stored in structured arrays**. Instead of providing the subsequent module with a sequence of independently extracted values, the VI provides the navigation information in a consistent and accessible arrangement.

These data are then available to the **Navigation Message processing stage** for further conversion and message generation.

3.5 PRN C/A Code Generator VI

The **PRN C/A Code Generator VI** is an important module in the developed GPS L1 signal-generation system. Its purpose is to generate the satellite-specific **GPS L1 Coarse/Acquisition (C/A) pseudo-random noise (PRN) code** required for the subsequent spreading and modulation stages.

In GPS, different satellites use different PRN sequences. Although the satellites transmit within the same L1 frequency band, the unique PRN sequence assigned to each satellite allows a receiver to distinguish and identify individual satellite signals through correlation. Therefore, generation of the correct PRN sequence is essential for producing a valid satellite-specific GPS L1 signal.

The developed implementation uses **LabVIEW as the main interface and Python as the backend for PRN generation**. The generated sequence is subsequently stored in a **CSV file**, providing an intermediate representation of the generated C/A code that can be accessed and processed within the overall LabVIEW-based system.

3.5.1 Background of PRN and C/A Code

A **Pseudo-Random Noise (PRN) code** is a deterministic binary sequence that has noise-like characteristics. Although the sequence appears random, it is generated according to a predefined mathematical procedure and can therefore be reproduced exactly when the same initial conditions and parameters are used.

PRN codes are fundamental to GPS because the system uses spread-spectrum signaling and allows multiple satellites to operate in the same frequency band. Each GPS satellite is assigned a unique PRN sequence. A receiver can generate the expected PRN code locally and correlate it with the received signal to determine whether the corresponding satellite signal is present.

The **Coarse/Acquisition (C/A) code** is the PRN spreading code associated with the GPS L1 C/A signal. It consists of **1023 chips** and has a chip rate of **1.023 Mcps**, resulting in a code period of approximately **1 ms**. The term *chip* is used for an individual element of a spreading code.

The C/A code is generated using two **10-stage Linear Feedback Shift Registers (LFSRs)**, commonly referred to as **G1 and G2**. The G1 sequence is common to the C/A-code generation process, while different combinations of G2 outputs are selected according to the PRN number. This produces a different C/A sequence for each satellite PRN.

Brief Historical Background: The GPS signal design uses spread-spectrum techniques to allow signals from multiple satellites to share the available frequency spectrum while remaining distinguishable by their individual codes. The C/A code became an important part of the original GPS L1 civilian signal because it provides both satellite identification and signal-spreading functionality.

The use of unique PRN sequences means that a GPS receiver does not need a separate frequency for every satellite. Instead, it can use the known PRN sequence associated with a satellite and perform correlation to acquire and track that satellite's signal.

For the present project, the important aspect is that the selected PRN determines the C/A sequence used in the GPS L1 signal-generation process.

3.5.2 Developed PRN C/A Code Generator VI

A dedicated **PRN C/A Code Generator VI** was developed to generate the required C/A sequence for the selected GPS satellite. The VI provides different operating modes for selecting the PRN and integrates the PRN-generation algorithm with the LabVIEW-based signal-generation system.

The developed VI supports three operating modes:

- **Default Mode**
- **Manual Mode**
- **Random Mode**

In **Default Mode**, the predefined PRN configuration is used for generating the C/A sequence.

In **Manual Mode**, the user specifies the required PRN number. The corresponding PRN configuration is then used by the backend program to generate the required sequence.

In **Random Mode**, a PRN is selected automatically from the available PRN configurations. This mode is useful during testing because different satellite-specific codes can be generated without manually entering a PRN number.

The final output for each selected PRN is a **1023-chip C/A code sequence**.

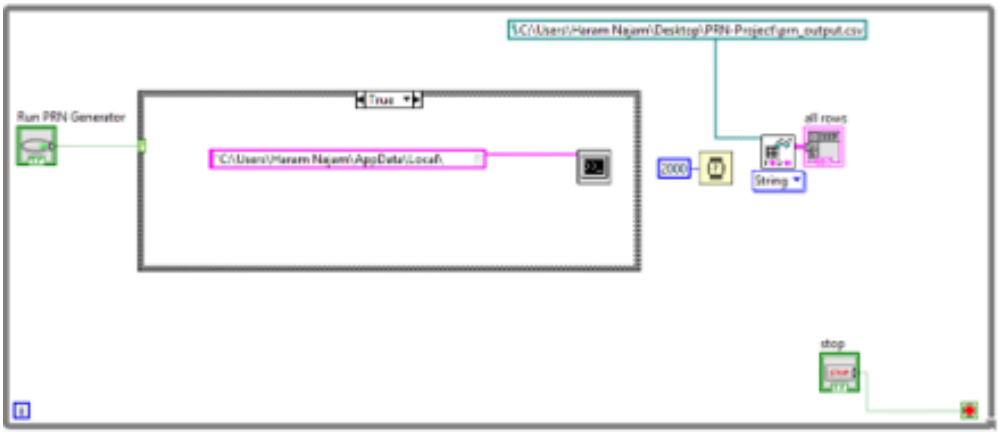


Figure 3.9: Complete PRN C/A Code VI implemented in LabVIEW.

3.5.3 LabVIEW–Python Implementation

Unlike a VI implemented entirely using LabVIEW functions, the PRN C/A Code Generator uses a **Python backend** for the actual code-generation operation. LabVIEW provides the main interface and integration with the remaining GPS signal-generation system, while Python performs the repetitive PRN-code generation algorithm.

The Python program contains the required **PRN-specific G2 tap configurations**. When a PRN number is selected, the corresponding tap configuration is obtained and used during the generation process.

The overall architecture of the developed implementation is shown in Figure 3.10.

This approach separates the ****user interface and system-level control**** from the computational PRN-generation algorithm while maintaining the generated code as an intermediate data file.

3.5.4 Working of the PRN C/A Code Generator VI

The operation of the developed PRN C/A Code Generator VI begins with the selection of the required operating mode.

Step 1: Select Operating Mode The user selects one of the three available modes:

- Default Mode

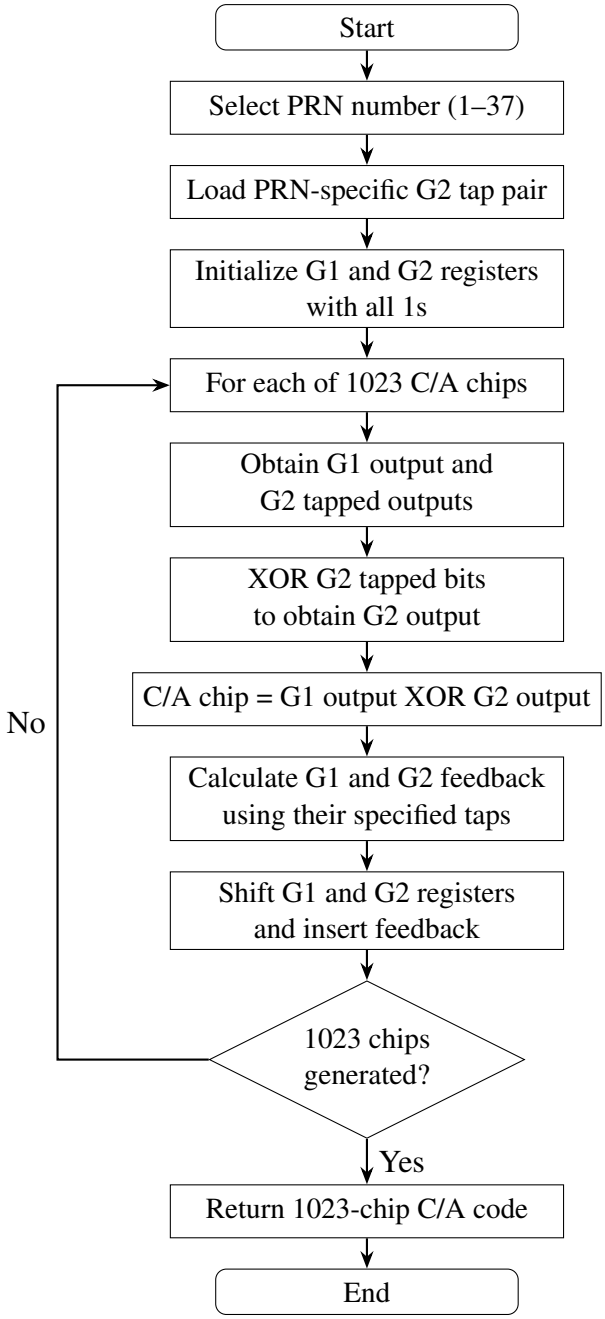


Figure 3.10: Flowchart of the GPS C/A PRN code generation process.

- Manual Mode
- Random Mode

The selected mode determines how the PRN number is obtained.

Step 2: Determine the PRN Number In Manual Mode, the user enters the required PRN number.

In Default Mode, the predefined PRN is selected.

In Random Mode, the backend selects a PRN automatically according to the available PRN configurations.

The selected PRN determines the specific C/A sequence that will be generated.

Step 3: Select PRN-Specific Tap Configuration The Python backend contains a predefined configuration of **G2 tap pairs** for the supported PRNs.

The selected PRN is used to identify the corresponding G2 tap positions. Since different PRNs use different tap configurations, changing the PRN results in a different C/A sequence.

Step 4: Initialize the G1 and G2 Registers The C/A-code generation process uses two 10-stage shift registers:

- **G1 register**
- **G2 register**

The registers are initialized before the generation process begins. The two registers are then updated repeatedly during the generation of each chip.

Step 5: Generate G1 and G2 Outputs At each iteration, the current states of the G1 and G2 registers are used to produce the required outputs.

The G2 output is determined using the tap positions associated with the selected PRN. The register states are then updated for the next iteration.

Step 6: Combine the Register Outputs The G1 output and selected G2 outputs are combined using the required binary XOR operation.

The generated C/A chip can therefore be represented conceptually as:

$$C/A\ Chip = G1\ Output \oplus Selected\ G2\ Outputs \tag{3.5.1}$$

This operation produces one chip of the satellite-specific C/A sequence.

Step 7: Generate 1023 Chips The generation process is repeated until **1023 chips** have been produced.

The sequence-generation process can be represented as:

Register Initialization $\rightarrow$ Generate Chip $\rightarrow$ Store Chip $\rightarrow$ Update Registers $\rightarrow$ Generate Next Chip

This process continues until the complete 1023-chip sequence has been generated.

Step 8: Store the Generated Code Once the 1023 chips have been generated, the Python backend stores the resulting sequence as numerical data and writes it to a **CSV file**.

The CSV file provides a simple representation of the generated PRN sequence and allows the generated data to be inspected and used by the LabVIEW processing environment.

3.5.5 Python Backend for PRN Generation

The Python program forms the computational backend of the developed PRN C/A Code Generator. Rather than placing the complete PRN-generation algorithm directly inside the LabVIEW block diagram, the algorithm is implemented in Python and invoked as part of the overall system.

The Python implementation contains the PRN-specific tap configuration and performs the register-based generation process. For each requested PRN, the program performs the following operations:

1. Obtains the PRN number.
2. Selects the corresponding G2 tap pair.
3. Initializes the G1 and G2 registers.
4. Generates the register outputs.
5. Combines the required outputs.
6. Stores each generated chip.
7. Repeats the process until 1023 chips are generated.
8. Saves the resulting sequence to a CSV file.

Appendix A – Python Source Code for PRN C/A Code Generation

CSV Generation

After the PRN C/A code has been generated, the Python backend stores the resulting sequence in a **CSV (Comma-Separated Values) file**.

The generated file contains the individual C/A-code values corresponding to the 1023 generated chips. The CSV format was used because it provides a simple and easily accessible representation of the generated sequence and allows the data to be inspected independently during testing.

The CSV file therefore acts as an intermediate representation between the Python-based PRN-generation process and the LabVIEW-based signal-processing environment.

3.5.6 Block Diagram of PRN C/A Code Generator VI

The block diagram of the developed PRN C/A Code Generator VI illustrates the internal processing performed by the module. The VI receives the selected PRN and operating mode, communicates with the Python backend for code generation, and handles the resulting PRN sequence for subsequent GPS signal processing.

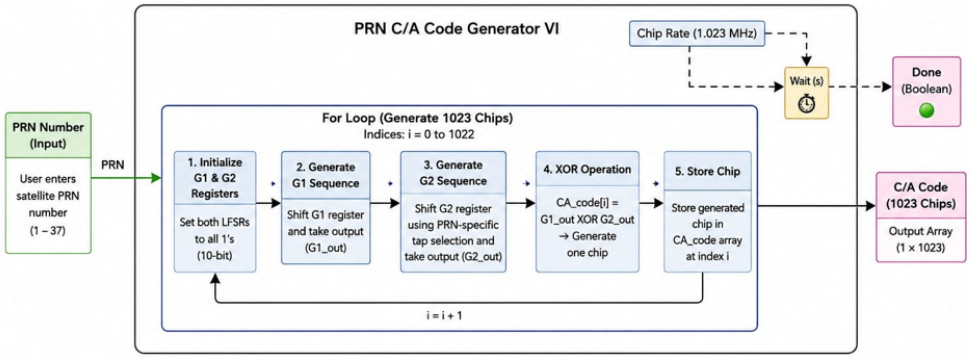


Figure 3.11: Block Diagram of PRN C/A Code Generation VI.

3.5.7 Processing Sequence

The main functions implemented in the PRN C/A Code Generator are summarized in Table 3.3.

Table 3.3: Main functions of the developed PRN C/A Code Generator.

Component/Stage	Implementation	Function
PRN Mode Selection	LabVIEW	Selects Default, Manual, or Random mode.
PRN Selection	LabVIEW/Python	Determines the required satellite PRN.
PRN Configuration	Python	Provides PRN-specific G2 tap positions.
G1 Register	Python	Generates the G1 sequence.
G2 Register	Python	Generates the PRN-dependent sequence.
Code Generation	Python	Produces the C/A chips.
Iteration	Python	Repeats generation until 1023 chips are produced.
Data Storage	Python/CSV	Stores the generated sequence.
Signal Integration	LabVIEW	Makes the PRN sequence available to subsequent stages.

3.5.8 Output of the PRN C/A Code Generator VI

The output of the developed VI is a **1023-chip PRN C/A sequence corresponding to the selected PRN**. The sequence can be observed as a binary waveform in LabVIEW, where the horizontal axis represents the chip index and the vertical axis represents the generated code value.

The generated PRN sequence is subsequently used as the satellite-specific spreading code in the GPS L1 signal-generation process. Therefore, the correct generation of the PRN sequence is essential before the navigation data can be combined with the spreading code and passed to the subsequent modulation stages.

3.6 Navigation Message Generator VI

The **Navigation Message Generator VI** is one of the key processing modules developed for the GPS L1 signal-generation system. Its purpose is to receive the navigation information in GPS-word form, convert the received word into individual navigation bits, and provide the resulting bit sequence to the subsequent GPS L1 signal-processing stages.

The navigation message contains satellite-related information required for GPS operation. In the developed system, the navigation data are obtained from the navigation-message processing stage and are transferred to the LabVIEW environment through a **TCP communication interface**. The LabVIEW VI then processes the received GPS word and converts it into the required Boolean navigation-bit representation.

The final implementation therefore uses a **TCP-based navigation-message conversion approach** rather than generating an arbitrary duration-based binary sequence directly inside LabVIEW.

3.6.1 Navigation Message in the GPS L1 Signal

The GPS L1 C/A signal consists of the carrier, the satellite-specific PRN C/A spreading code, and the navigation data. The navigation data provide the information associated with the transmitting satellite and are subsequently combined with the PRN code during the L1 signal-generation process.

In the developed project, the navigation-message processing stage is positioned after the RINEX-based data-processing modules. The RINEX File Parser extracts the required satellite navigation information, while the GPS Data Sorting VI organizes the extracted parameters. The navigation-message stage then provides the navigation-word and bit data required for integration with the remaining L1 signal-generation modules.

3.6.2 TCP-Based Navigation Message Approach

The final implementation uses **TCP communication** to transfer navigation words to LabVIEW.

Instead of performing the complete navigation-message processing directly inside the LabVIEW block diagram, the navigation-message data are supplied through a TCP connection. This provides a separate processing source for the navigation words and allows LabVIEW to concentrate on receiving and converting the data for use within the GPS signal-generation chain.

The basic architecture of the developed implementation is shown in Figure 3.12.

This approach was adopted as the final implementation after the initial development work on navigation-bit generation.

3.6.3 Developed Navigation Message Generator VI

The developed Navigation Message Generator VI provides the LabVIEW-side interface for receiving and converting the navigation information.

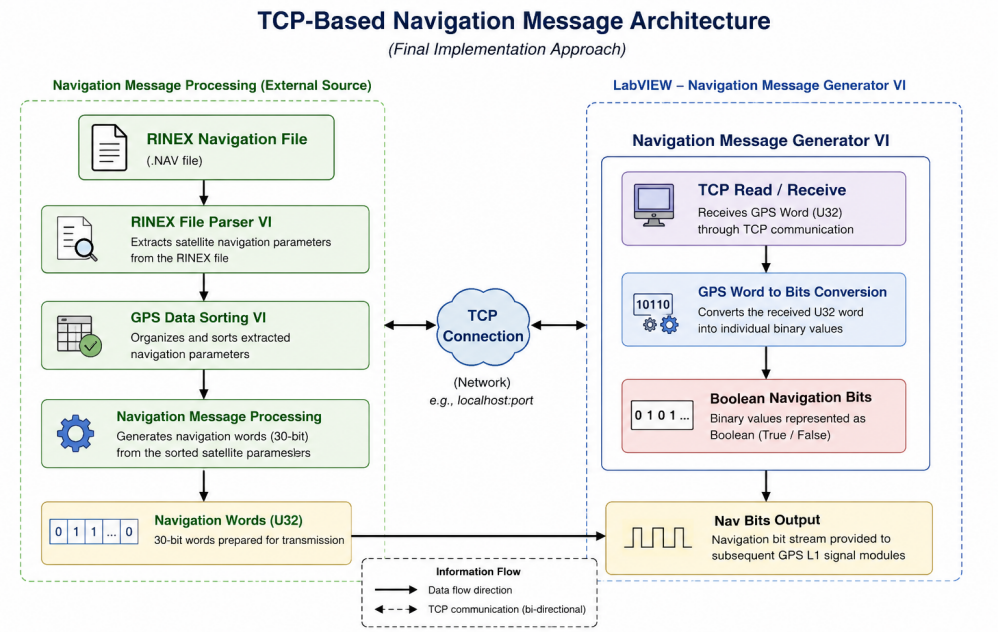


Figure: TCP-Based Navigation Message Architecture for GPS L1 Signal Generation

Figure 3.12: TCP Based Navigation Message Architecture for GPS L1 Signal Generation.

The VI receives a **GPS Word represented as a U32 value** through the TCP communication process. The received numerical word is then converted into its individual binary values. These values are represented as Boolean navigation bits and provided through the **Nav Bits** output.

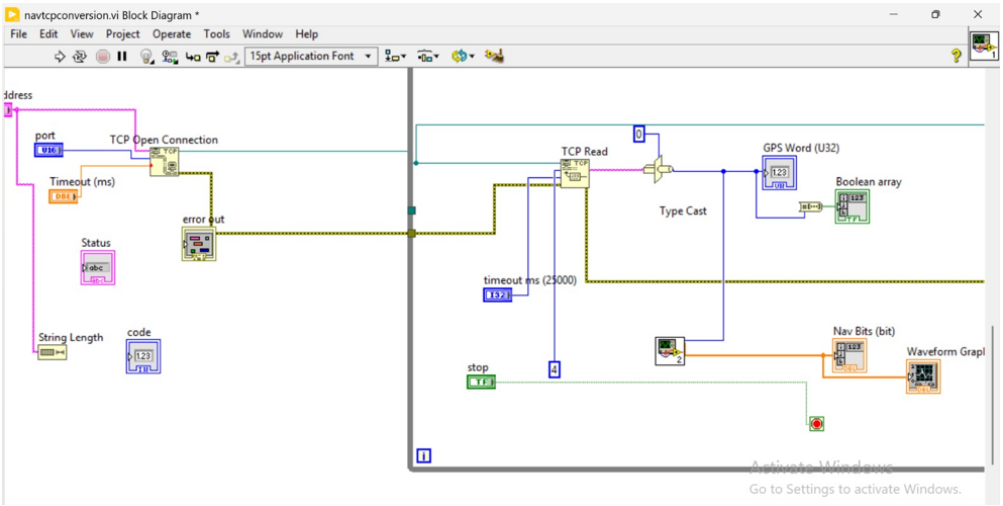


Figure 3.13: TCP-based Navigation Message VI implemented in LabVIEW.

The figure illustrates the overall arrangement of the developed VI before the individual processing stages are discussed.

3.6.4 Working of the Navigation Message Generator VI

The operation of the developed VI is described below from the input stage to the final navigation-bit output.

Step 1: Navigation Message Source The navigation-message source provides the required GPS navigation words. These words represent the navigation information that is supplied to the LabVIEW-based GPS L1 signal-generation system.

The navigation data are prepared externally and made available to the LabVIEW VI through the TCP communication interface.

Step 2: Establish TCP Communication The LabVIEW VI establishes communication with the navigation-message source through a **TCP connection**.

The TCP interface acts as the communication link between the navigation-message source and LabVIEW. Once the connection is available, navigation words can be transferred to the VI for processing.

This separation makes the navigation-message processing stage modular and avoids placing the complete external navigation-data handling process directly inside the main LabVIEW signal-generation block diagram.

Step 3: Receive the GPS Word After the TCP connection is established, the VI receives the navigation information in the form of a **GPS Word**.

Within LabVIEW, the received word is represented as a **U32 numerical value**. This U32 representation allows the complete received word to be handled as a single numerical quantity before its individual binary components are extracted.

Step 4: Convert the GPS Word into Bits The received U32 GPS Word is passed to the bit-conversion stage. The purpose of this stage is to convert the numerical representation of the word into its individual binary values.

The conversion changes the data from a numerical GPS-word representation into individual Boolean bits:

$$\text{GPS Word} \rightarrow \text{U32 Representation} \rightarrow \text{Individual Boolean Bits} \tag{3.6.1}$$

This conversion is required because the subsequent GPS signal-generation process operates on individual navigation bits.

Step 5: Generate the Nav Bits Output After conversion, the individual binary values are arranged into the **Nav Bits** output.

The `Nav Bits` output represents the navigation information in Boolean form and provides the required interface for subsequent GPS signal-processing operations.

The navigation bits can also be monitored within the VI to verify that the received word is being converted into the expected binary representation.

Step 6: Display and Monitor the Navigation Bits The developed VI includes a display for observing the generated navigation-bit sequence.

The waveform representation allows the operation of the conversion stage to be monitored during testing. This is useful for checking whether data are being received correctly through TCP and whether the received GPS word is being converted into a binary sequence.

Step 7: Continue Processing Navigation Words The process is repeated as additional navigation words are received.

Therefore, the final VI is not simply a one-time conversion block. It forms part of the navigation-data stream between the external navigation-message source and the LabVIEW GPS L1 signal-processing chain.

The complete conversion process can be summarized as:

3.6.5 TCP-Based Navigation Message Architecture

The TCP-based implementation separates the navigation-message source from the LabVIEW signal-processing environment. The TCP connection provides the communication interface, while the LabVIEW VI performs the required word-to-bit conversion.

3.6.6 Navigation Data Rate

The GPS navigation data used in the L1 C/A signal have a nominal data rate of **50 bits per second**. This timing is important when the navigation data are eventually combined with the PRN C/A code and integrated into the GPS L1 signal.

The Navigation Message Generator therefore forms an important timing-related stage between the navigation-data source and the subsequent signal-generation modules.

The detailed timing measurements and experimental output of the implemented navigation-message system are discussed separately in **Chapter 4: Results and Discussion**.

3.6.7 Block Diagram of the Navigation Message Generator VI

The block diagram of the developed Navigation Message Generator VI illustrates the internal processing performed by the module. The VI receives the navigation data through the TCP interface, represents the received information as a GPS Word/U32 value, performs the required bit conversion, and provides the resulting Boolean values as the `Nav Bits` output.

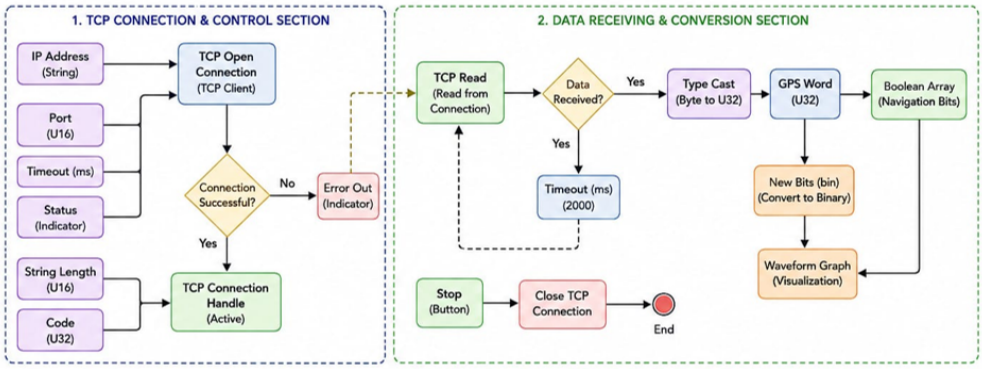


Figure 3.14: Block Diagram of Navigation Message VI.

The block diagram can be explained from left to right. The TCP section receives the navigation data, the received information is represented as a GPS Word/U32 value, and the conversion section extracts the individual binary values. These values are then provided as the `Nav Bits` output.

3.6.8 Main Functions of the Navigation Message Generator VI

The main processing stages of the Navigation Message Generator VI are summarized in Table 3.4.

Table 3.4: Main processing stages of the Navigation Message Generator VI.

Stage	Component/Operation	Function
1	Navigation message source	Provides the required GPS navigation words.
2	TCP connection	Provides communication between the navigation-message source and LabVIEW.
3	GPS Word	Represents the received navigation word for processing.
4	U32 representation	Holds the received GPS word as a numerical value.
5	Bit conversion	Converts the numerical word into individual binary values.
6	Boolean Nav Bits	Represents the navigation information in Boolean form.
7	Waveform/display	Allows the converted navigation-bit sequence to be monitored.
8	Output	Provides navigation bits to the subsequent GPS L1 signal-processing stage.

3.6.9 Output of the Navigation Message Generator VI

The output of the developed Navigation Message Generator VI is the **Nav Bits** sequence obtained by converting the received GPS navigation words into individual Boolean values.

The output represents the navigation information in the binary form required by the subsequent GPS L1 signal-generation stages. The waveform and display section of the VI also provides a visual representation of the generated navigation-bit sequence, which can be used to monitor the operation of the TCP reception and conversion process.

3.7 Carrier Signal Generation VI

The Carrier Signal Generation VI is a part of the GPS L1 signal-generation system that is created. The job of this part is to make the carrier waveform that is needed for the GPS L1 signal-processing process. The carrier that is made gives a wave that looks like a wave. This wave is used to mix with the navigation and PRN C/A parts during the modulating part.

In the system that is created the part that makes the carrier is treated as its separate part. This way of working makes it possible to make the carrier wave look at it and check it on its own before it is given to the part of the signal-processing work.

3.7.1 Role of the Carrier in GPS L1 Signal Generation

The GPS L1 C/A signal uses a carrier frequency of about

1575.42 MHz. The navigation data and PRN C/A code are the parts that have the information and the spreading part of the signal. The carrier is the wave that has the shape of a wave. This is needed to create the L1 signal.

The part that makes the carrier is one of the parts needed to build the final GPS L1 wave.

3.7.2 Carrier Parameters

The Carrier Signal Generation VI that is created makes a wave that looks like a sine wave. This is done based on the carrier parameters that are needed. The main things that are looked at when making the carrier are:

- carrier frequency;
- sampling frequency;
- carrier amplitude;
- phase; and
- number of samples or how long the signal is.

For the GPS L1 signal the carrier frequency that is used as the example is

$$f_c = 1575.42\,\text{MHz} \tag{3.7.1}$$

The sampling choices that are made decide how the carrier is shown as a signal in the LabVIEW system.

3.7.3 Mathematical Representation of the Carrier

A sine wave carrier can be shown in time by

The carrier signal is represented as a sequence of discrete samples. If f_s is the sampling frequency and n is the sample index, the carrier samples can be expressed as:

$$s_c[n] = A\cos\left(2\pi f_c \frac{n}{f_s}\right) \tag{3.7.2}$$

where A is the carrier amplitude, f_c is the carrier frequency, f_s is the sampling frequency, and n is the sample index.

$$c[n] = A\cos\left(2\pi \frac{f_c}{f_s} n + \phi\right) \tag{3.7.3}$$

The samples that come out make the version of the sine wave carrier inside the LabVIEW system.

3.7.4 Developed Carrier Signal Generation VI

A special Carrier Signal Generation VI was made to make the carrier wave needed for the process of making the GPS L1 signal. The VI takes in the information that is needed to describe the carrier and makes the digital wave.

The VI that is made is shown before the processing explanation. This helps the way of working to be understood before looking at the parts one by one.

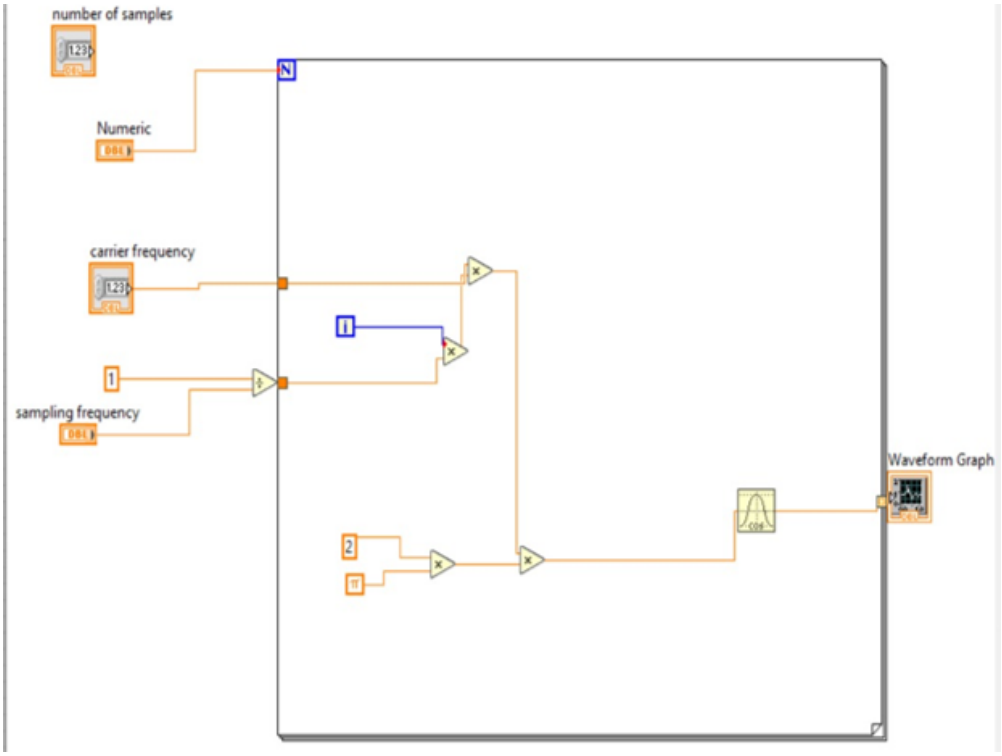


Figure 3.15: Carrier Signal Generation VI in LabView.

3.7.5 Working of the Carrier Signal Generation VI

The way the VI that is made works can be split into the following steps.

Step 1: Set Carrier Parameters The VI gets the information that is needed to make the carrier. These details decide the frequency the way it is sampled the size the beginning point and how many samples there are in the wave that is made.

Step 2: Make the Sample Positions The frequency at which it is sampled decides how far apart each digital sample is. The time for the nth sample is given by

$$t_n = \frac{n}{f_s} \tag{3.7.4}$$

where n is the sample index and f_s is the sampling frequency.
These positions are used to find the carrier value for each one.

Step 3: Make the Sine Wave Carrier For every position where a sample is taken the VI makes the value of the carrier using the equation for the sine wave. The numbers that come out are parts of the carrier wave that is needed.

Step 4: Make the Carrier Wave The samples that are made are put into a list or a wave picture inside LabVIEW. This creates the digital form of the carrier signal.

Step 5: Show and Check the Wave The wave that is made can be shown using a graph or a wave picture in LabVIEW. This makes it possible to look at the signal and check that it is correct, in the middle of the making process.

Step 6: Give the Carrier to the Next Step After the wave is made and checked the carrier wave is given to the part of the process. There it is used with the processed GPS info during the making of the L1 signal.

3.7.6 Block Diagram of the Carrier Signal Generation VI

The block diagram of the Carrier Signal Generation VI that is made shows how the processing works from the carrier parameters to the wave that is made.

The block diagram is read from the part where the parameters come in to the part where the wave comes out. The parameters that are put in describe the carrier that is needed. The carrier samples are made based on the sampling choices that are made. The samples that come out are put into the wave that is given out.

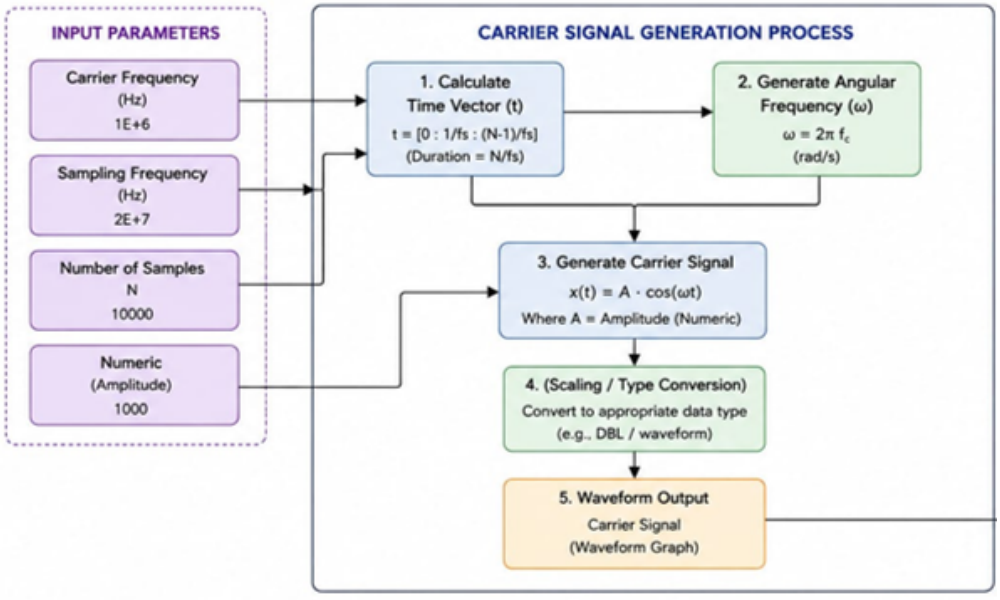


Figure 3.16: Block Diagram Of Carrier Signal Generation VI.

3.7.7 Processing Sequence

The principal processing functions of the Carrier Signal Generation VI are summarized in Table 3.5.

Table 3.5: Main processing functions of the Carrier Signal Generation VI.

Stage	Component/Operation	Function
1	Carrier frequency	Defines the required L1 carrier frequency.
2	Sampling frequency	Defines the digital sampling rate used to represent the carrier.
3	Amplitude	Defines the magnitude of the generated carrier waveform.
4	Phase	Defines the initial phase of the carrier.
5	Sample generation	Produces the digital sample positions required for waveform generation.
6	Sinusoidal calculation	Calculates the individual carrier samples.
7	Waveform formation	Arranges the generated samples into the carrier waveform.
8	Display and verification	Provides a visual representation of the generated carrier for inspection.
9	Carrier output	Provides the generated carrier to the subsequent signal-processing stage.

3.7.8 Output of the Carrier Signal Generation VI

The output of the developed Carrier Signal Generation VI is a **digitally represented sinusoidal carrier waveform** corresponding to the specified carrier parameters.

The generated waveform can be observed in LabVIEW using the corresponding graph or waveform indicator. The waveform provides an intermediate verification point for confirming the operation of the carrier-generation stage before integration with the remaining GPS L1 signal-processing modules.

3.7.9 Integration with the GPS L1 Signal-Generation System

The Carrier Signal Generation VI forms one of the major signal-building modules of the developed GPS L1 signal-generation system. It operates alongside the navigation-message and PRN C/A code processing modules.

The carrier-generation module therefore functions as a dedicated signal-generation stage and does not generate the navigation data or PRN C/A code. Its output is intended to be integrated with the processed GPS information during the subsequent stages of the L1 signal-generation process.

3.8 Final GPS Signal Integration Using NI GPS Data Streaming VI

The development of the GPS L1 signal generation system was done using a modular signal processing approach. The entire signal generation process was split into functional steps and special LabVIEW VIs were created for the main processing tasks.

The developed modules included the RINEX File Parser VI, GPS Data Sorting VI, Navigation Message Generator VI, PRN C/A Code Generator VI spreading stage, BPSK Modulation VI and Carrier Signal Generation VI. This modular setup made it possible to implement and test each GPS processing concept on its own before moving with full system integration.

The next step, in the development involved connecting the generated GPS signal to the USRP-2920 for hardware based signal generation and sending. The USRP-2920 works with the LabVIEW environment through the National Instruments USRP interface. The overall system design therefore needs both GPS signal processing functions and specific hardware interface parts.

At this point the project used the NI GPS Data Streaming VI, a built-in National Instruments version that offers full GPS signal generation and hardware streaming features. This method allowed each of the individual GPS processing steps to be looked at within a SDR transmission setup.

3.8.1 Need for the Integrated NI Implementation

The individually developed VIs provide the fundamental GPS signal-processing chain.

Generating the signal-processing output is only one part of an SDR transmission system. For transmission, the generated digital samples must also be transferred to the SDR hardware using the appropriate hardware interface.

The project hardware architecture uses the **USRP-2920**, with the computer and USRP connected through the required communication interface.

The USRP receives the generated signal data and performs the required RF signal generation for the transmission stage.

The required NI-USRP functionality is provided through specialized National Instruments components. These components form part of the National Instruments software implementation. Therefore, the NI GPS Data Streaming VI was used as the integrated hardware-streaming implementation.

Thus, the use of the NI GPS Data Streaming VI represents the **hardware-integration stage of the project** rather than a replacement for the individually developed GPS processing modules.

3.8.2 NI GPS Data Streaming VI

The **NI GPS Data Streaming VI** is a built-in National Instruments VI that integrates GPS signal generation with the data-streaming functionality required for SDR operation.

Because the complete block diagram is large, its operation is described in terms of its major functional sections rather than individual LabVIEW blocks. The major sections are:

1. GPS configuration;
2. USRP transmission configuration;
3. GPS data generation;
4. Producer loop;

5. GPS data queue;
6. Consumer loop;
7. waveform monitoring;
8. RF signal generation and streaming; and
9. shutdown and queue destruction.

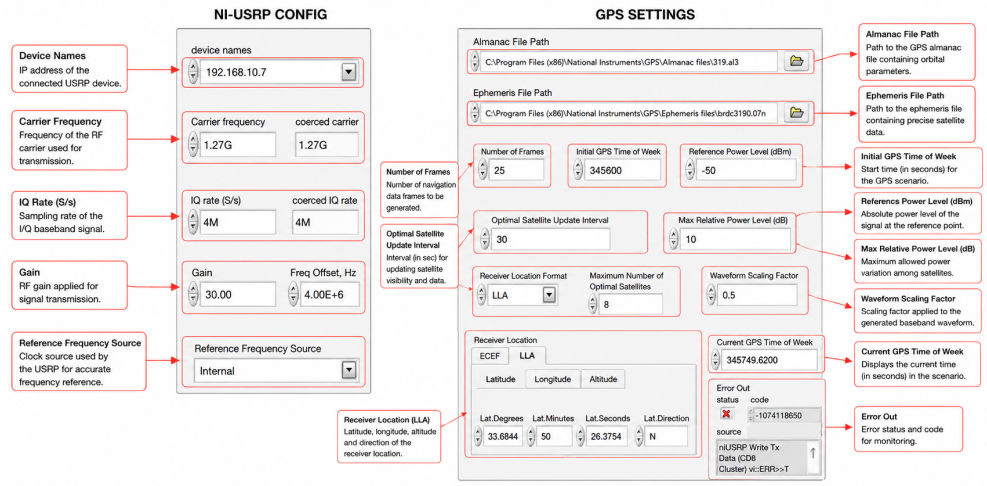


Figure 3.17: Front panel of the NI GPS Data Streaming VI showing GPS and USRP configuration and generation controls.

3.8.3 Relationship Between the Developed VIs and the NI GPS Data Streaming VI

The NI GPS Data Streaming VI contains functions that are conceptually related to several processing stages developed during the project. This relationship demonstrates how the modular development corresponds to the functional stages of an integrated GPS implementation.

The comparison demonstrates that the NI VI is used as the integrated implementation for the final streaming stage. It is therefore considered part of the overall hardware-integration architecture rather than an alternative description of the individual modules developed during the project.

3.8.4 Overall Working of the NI GPS Data Streaming VI

The complete operation of the NI GPS Data Streaming VI can be divided into configuration, waveform generation, buffering, streaming, monitoring, and shutdown operations.

3.8.5 GPS Configuration Stage

The first major section of the NI GPS Data Streaming VI is the **GPS Configuration stage**. This section establishes the conditions under which the GPS waveform is generated.

Table 3.6: Relationship between the developed GPS processing modules and the integrated NI GPS Data Streaming VI.

Developed Module	**Related Integrated Function**	**Purpose**
RINEX File Parser VI	Navigation/ephemeris data input	Provides satellite navigation information.
GPS Data Sorting VI	GPS parameter organization	Organizes satellite parameters for subsequent processing.
Navigation Message Generator VI	Navigation-data generation	Provides navigation information for signal generation.
PRN C/A Code Generator VI	PRN/code generation	Produces satellite-specific spreading information.
Spreading stage	GPS signal generation	Combines navigation information with the PRN sequence.
BPSK Modulation VI	Signal modulation	Produces the modulated GPS signal.
Carrier Signal Generation VI	Carrier generation	Provides the required L1 carrier.
NI GPS Data Streaming VI	**Complete integrated implementation**	**Combines GPS waveform generation with continuous data streaming and hardware interface functionality.**

The configuration includes parameters such as:

- receiver location;
- almanac data;
- ephemeris data;
- GPS time;
- satellite power levels;
- number of satellites; and
- waveform scaling.

These parameters define the GPS scenario represented by the generated waveform. For example, the receiver location defines the geographical scenario, while ephemeris and almanac information provide satellite-related parameters. GPS time establishes the time reference, and the satellite power levels define the relative signal strengths used in the generated scenario.

3.8.6 USRP Transmission Configuration

After the GPS scenario is configured, the transmission parameters required for the USRP are established.

The main transmission parameters include:

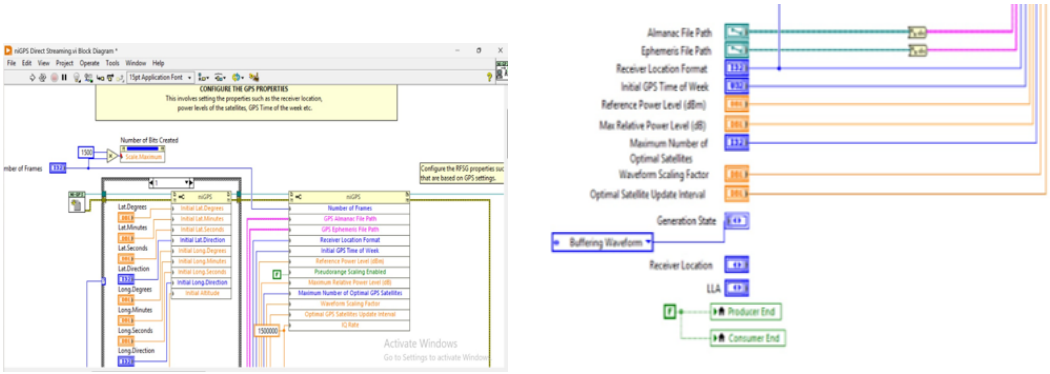


Figure 3.18: GPS configuration section of the NI GPS Data Streaming VI.

- IQ rate;
- gain;
- carrier frequency;
- frequency offset;
- device identification; and
- reference frequency source.

These parameters establish the conditions under which the generated digital waveform is interfaced with the SDR hardware.

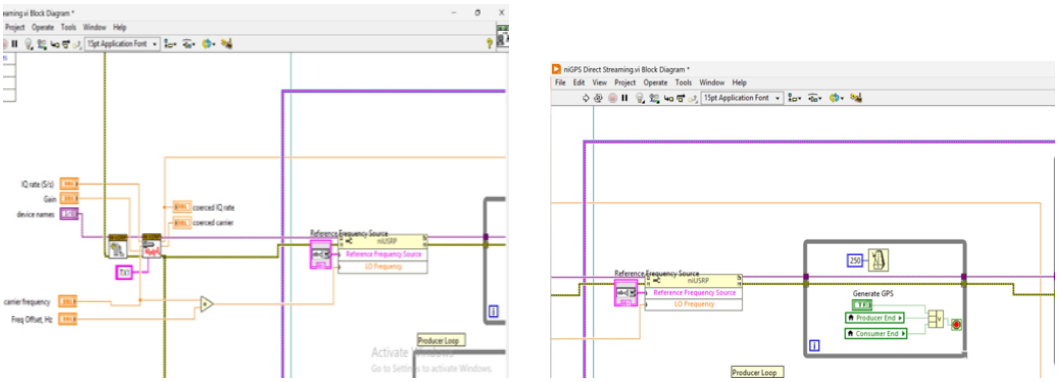


Figure 3.19: USRP transmission configuration section showing device, carrier frequency, IQ rate, gain, frequency offset, and reference frequency settings.

3.8.7 GPS Data Generation Loop

After initialization, the NI GPS VI enters the GPS data generation loop. The purpose of this loop is to continuously generate blocks of GPS waveform data.

The generation loop continuously produces GPS waveform data and places each generated block into the GPS data queue. Continuous generation is required because an SDR transmitter requires a stream of signal samples rather than a single isolated waveform calculation.

3.8.8 Producer–Consumer Architecture

An important architectural feature of the NI GPS Data Streaming VI is the use of a **Producer–Consumer architecture**. The architecture separates waveform generation from waveform transfer to the hardware.

The Producer generates and prepares GPS waveform blocks, while the Consumer retrieves these blocks and transfers them toward the hardware interface. A queue provides the communication mechanism between the two loops.

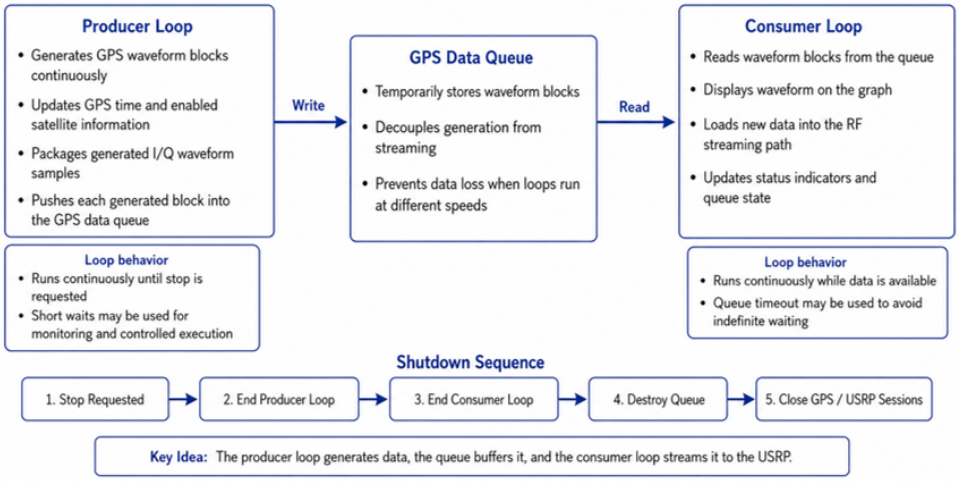


Figure 3.20: Producer–Consumer architecture used for continuous GPS waveform generation and streaming.

3.8.9 Purpose of the Producer–Consumer Architecture

Continuous SDR signal generation involves two major operations:

1. computational processing required to generate the GPS waveform; and
2. data transfer required to stream the generated waveform to the hardware.

These operations have different processing requirements and timing characteristics. Separating them into Producer and Consumer loops allows the generation and transmission processes to operate through a buffered communication path.

The architecture can be represented by the relationship

$$\text{Producer} \rightarrow \text{GPS Data Queue} \rightarrow \text{Consumer} \tag{3.8.1}$$

The queue acts as an intermediate storage mechanism between waveform generation and hardware streaming.

3.8.10 Producer Loop Working

The Producer Loop is responsible for generating GPS waveform blocks and placing them into the queue.

Its operation can be summarized as follows:

1. receive the configured GPS parameters;

2. generate the GPS waveform;
3. organize the generated samples into a data block;
4. place the generated block into the GPS data queue; and
5. continue generating subsequent blocks.

The processing relationship is

$$\text{GPS Parameters} \rightarrow \text{GPS Waveform Generation} \rightarrow \text{Data Block} \rightarrow \text{GPS Data Queue}. \tag{3.8.2}$$

The Producer Loop therefore represents the supply side of the continuous data stream.

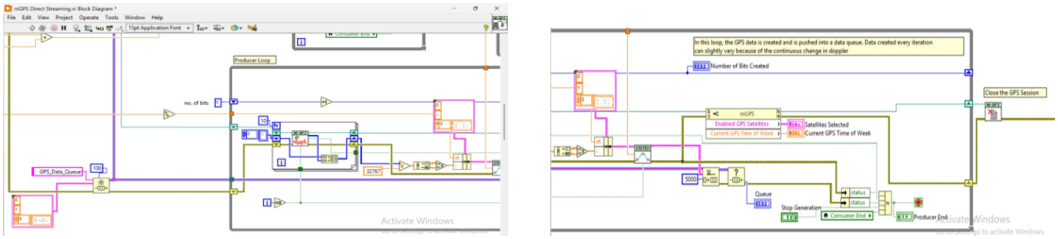


Figure 3.21: Producer loop responsible for generating GPS waveform blocks and placing them into the GPS data queue.

3.8.11 GPS Data Queue

The GPS data queue provides the intermediate communication mechanism between the Producer and Consumer loops. The Producer places generated waveform blocks into the queue, while the Consumer retrieves them for subsequent processing and streaming. The queue provides a degree of decoupling between waveform generation and hardware transmission. It also allows the system to process blocks of samples, which is suitable for continuous waveform streaming.

3.8.12 Consumer Loop Working

The Consumer Loop receives waveform blocks generated by the Producer Loop. Its main functions are:

1. retrieve a waveform block from the queue;
2. process or display the received data as required;
3. transfer the data toward the RF signal-generation interface; and
4. continue retrieving subsequent waveform blocks.

The simplified data path is represented by

$$\text{GPS Data Queue} \rightarrow \text{Consumer Loop} \rightarrow \text{Waveform Monitoring} \rightarrow \text{RF Signal Generator}. \tag{3.8.3}$$

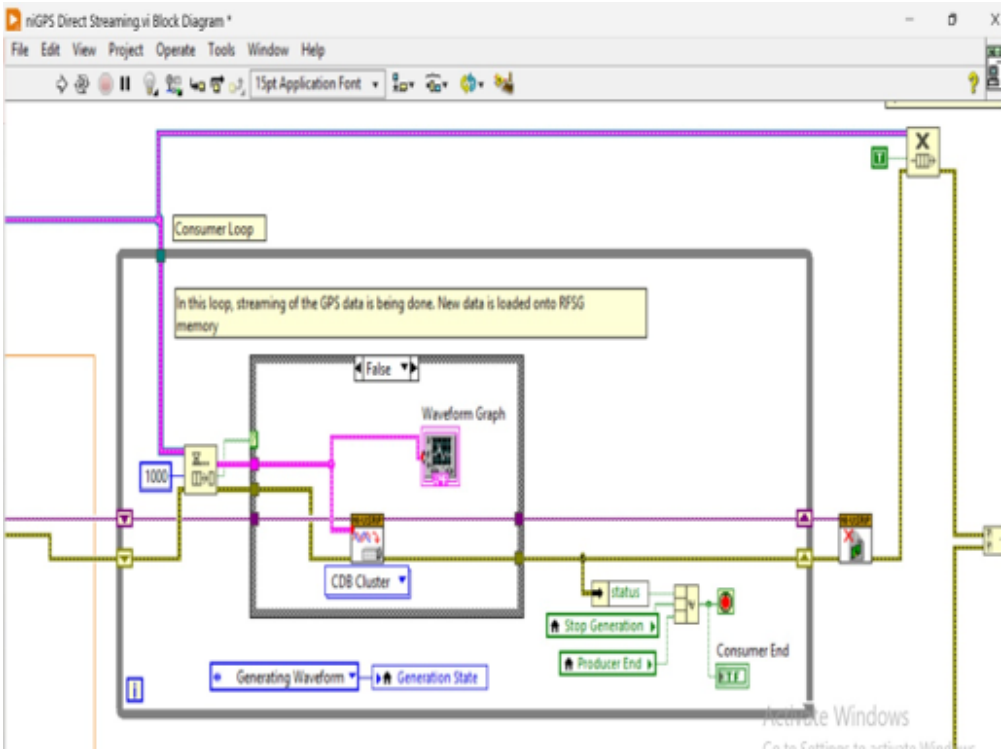


Figure 3.22: Consumer loop for retrieving generated GPS waveform blocks and transferring them toward the RF signal-generation interface.

3.8.13 Waveform Monitoring

The Consumer section also provides waveform monitoring functionality. The generated GPS waveform data can be displayed before or during streaming. This provides an intermediate verification point between GPS waveform generation and hardware transmission.

The monitoring path can be summarized as

$$\text{Generated GPS Data} \rightarrow \text{GPS Data Queue} \rightarrow \text{Consumer Loop} \rightarrow \text{Waveform Display} \rightarrow \text{Streaming}. \tag{3.8.4}$$

This allows the generated waveform to be observed as part of the integrated system.

3.8.14 RF Signal Streaming

After retrieving the generated waveform blocks, the Consumer Loop transfers the data toward the RF signal-generation and hardware interface.

3.8.15 Generation Control and System Monitoring

The NI GPS Data Streaming VI also includes control and monitoring functions during continuous operation. The final implementation monitors parameters associated with the GPS generation and streaming process, including:

- number of generated bits;
- selected satellites;

- GPS time;
- queue status;
- system state;
- stop condition; and
- error status.

These monitoring functions provide information about the state of the continuous generation and data-transfer process.

3.8.16 Shutdown and Queue Destruction

The final stage of the NI GPS Data Streaming VI is the controlled shutdown of the system. When a stop condition is detected, GPS generation is stopped and the resources associated with the streaming process are released.

Controlled shutdown prevents resources such as queues and hardware sessions from remaining active after the signal-generation process has ended.

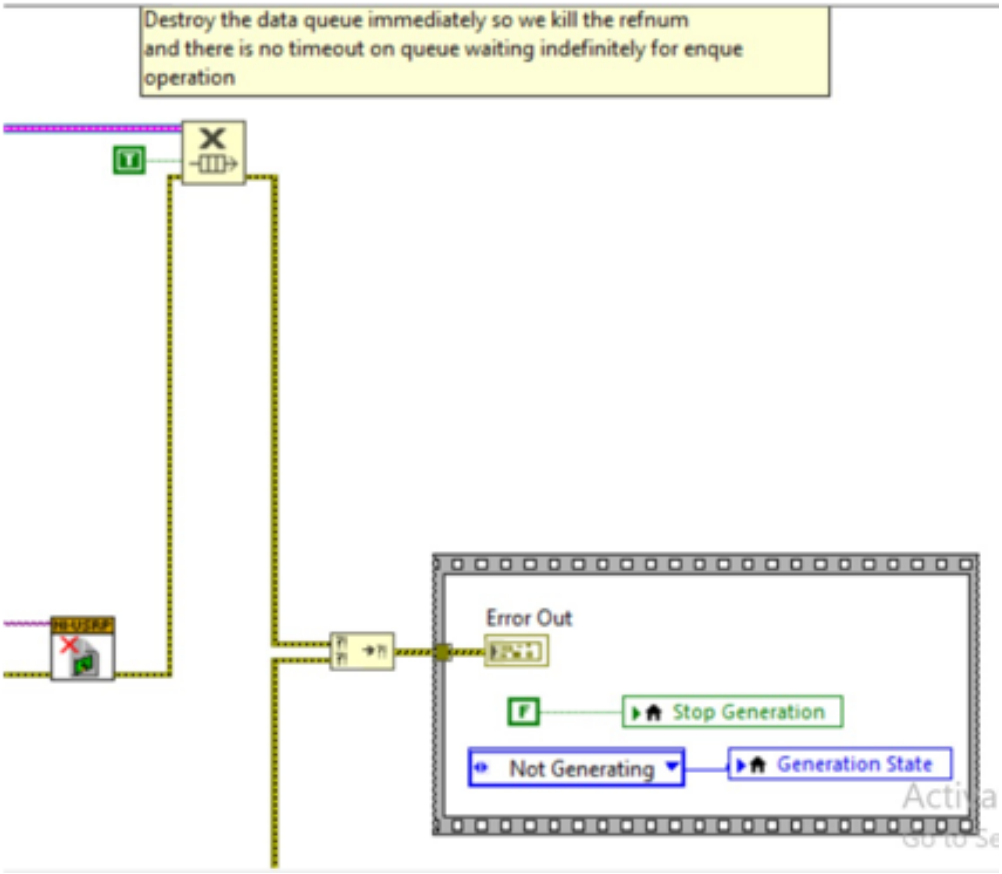


Figure 3.23: Queue Destruction and Shutdown destroys the data queue, stops signal generation, updates the system state, and reports any errors.

3.8.17 Comparison with the Developed Modular VIs

The comparison between the modular VIs and the integrated NI implementation demonstrates the continuity between the development and final integration stages.

Table 3.7: Comparison between modular GPS processing and integrated GPS streaming functions.

Function	**Developed Module/Stage**	**Integrated Streaming Function**
Navigation data	RINEX File Parser and Data Sorting	GPS configuration and navigation-data handling
Navigation message	Navigation Message Generator	Integrated GPS waveform generation
Satellite identification	PRN C/A Code Generator	Integrated GPS signal generation
Spreading	Developed spreading stage	Integrated signal generation
BPSK	BPSK Modulation VI	Integrated signal generation
Carrier	Carrier Signal Generation VI	Integrated GPS waveform generation
Sample generation	Individual waveform outputs	Continuous waveform generation
Data buffering	Module-level processing	GPS Data Queue
Parallel processing	Individual VIs	Producer–Consumer loops
Hardware streaming	Signal-processing output	RF signal generator and NI-USRP interface

3.8.18 Modular Development and Integrated Streaming

The development and integration stages can be distinguished according to their respective objectives.

The modular architecture emphasizes the development and verification of individual GPS signal-processing functions, whereas the integrated architecture emphasizes continuous waveform generation, buffering, and SDR transmission.

3.8.19 Complete Working Flow of the Final Integration

The complete final integration combines GPS scenario configuration, waveform generation, buffered data transfer, hardware streaming, and controlled shutdown.

3.8.20 Significance of the Integrated NI GPS VI

The use of the NI GPS Data Streaming VI provided the final integration layer required to move from individual GPS signal-processing modules to a complete SDR-oriented implementation.

The modular VIs developed during the project established the individual processing stages and their relationships. The integrated NI implementation then provided the architecture required for continuous waveform generation, buffering, and hardware streaming.

In particular, the Producer–Consumer architecture separates waveform generation from waveform transfer. The Producer generates GPS waveform blocks and places them into a queue, while the Consumer retrieves these blocks and transfers them toward the RF signal-generation and USRP interface.

The final implementation can therefore be viewed as the combination of two complementary stages:

$$\text{Modular GPS Signal Processing} + \text{Integrated SDR Streaming} \rightarrow \text{Complete GPS L1 System Architecture.} \tag{3.8.5}$$

3.9 Final Hardware Integration and GPS L1 Transmission

Following the development and integration of the individual GPS signal-processing stages, the complete GPS L1 signal-generation system was configured for hardware-based transmission using the **National Instruments USRP-2920**. The USRP-2920 served as the software-defined radio platform for transferring the generated GPS L1 waveform from the LabVIEW environment to the RF transmission chain.

For the final integrated implementation, the **NI GPS Data Streaming VI** was used to provide the GPS waveform generation and continuous data-streaming functionality required for communication with the USRP-2920. The internal architecture and operation of this VI have been discussed in the preceding sections.

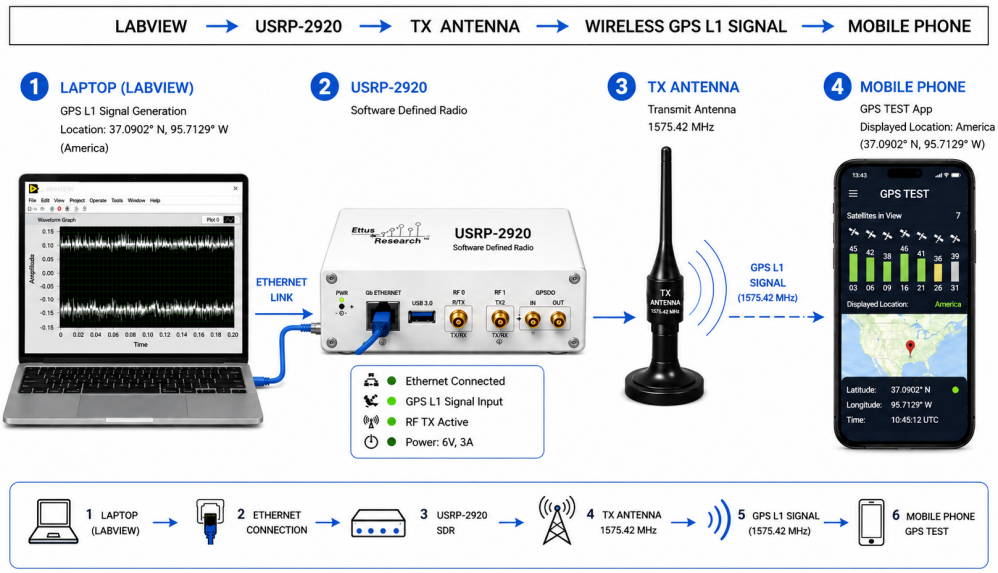


Figure 3.24: Overall experimental setup for transmission of the LabVIEW-generated GPS L1 signal through the USRP-2920 and transmit antenna to a mobile phone.

As illustrated in Figure 3.24, the GPS L1 signal-generation process begins in the **LabVIEW environment**, where the required GPS scenario and signal-generation parameters are configured. The generated signal is transferred through an **Ethernet connection** to the USRP-2920. The USRP then provides the RF transmission interface through the connected transmit antenna.

The transmitted signal corresponds to the **GPS L1 carrier frequency of 1575.42 MHz**. The RF signal is received wirelessly by a GPS-enabled mobile device, where a GPS testing application is used to observe the received satellite information and positioning result.

The complete signal path can be represented as:

$$\begin{aligned} &\text{LabVIEW} \rightarrow \text{Ethernet} \rightarrow \text{USRP-2920} \\ &\qquad \rightarrow \text{TX Antenna} \rightarrow \text{GPS L1 Signal} \rightarrow \text{GPS Receiver} \end{aligned} \tag{3.9.1}$$

The transmission was conducted as a **controlled laboratory experiment** for validation of the developed GPS L1 signal-generation system. The GPS scenario, receiver coordinates, satellite information, and other required parameters were configured in the LabVIEW environment before initiating the transmission.

At the receiver side, the transmitted signal was detected by the mobile GPS receiver. Multiple satellites were displayed by the GPS testing application, and the receiver achieved a **3D position fix**. The displayed position corresponded to the coordinates configured for the GPS scenario, providing receiver-level evidence of successful L1 signal generation and transmission.

The final experimental arrangement therefore demonstrates the complete transition from **software-based GPS signal generation to SDR-based RF transmission and receiver-level signal validation**.

3.10 Development and Verification Approach

The development of the GPS signal-generation system was carried out progressively because the complete implementation involved several interconnected processing stages, including GPS navigation-data processing, digital signal generation, LabVIEW implementation, and SDR integration.

A **modular development approach** was adopted in which the major processing functions were implemented and examined individually before being incorporated into the integrated system. This approach provided a structured method for verifying intermediate outputs and identifying issues at the relevant processing stage.

Verification was performed by observing the expected numerical and waveform outputs of the individual processing stages. The RINEX-processing stages were examined through their extracted navigation information, while the signal-generation stages were evaluated through their corresponding waveform outputs. After the individual stages were verified, they were integrated into the complete GPS signal-generation architecture.

The overall development progression can be summarized as:

This progressive methodology allowed the project to move systematically from individual signal-processing functions toward complete hardware-level operation.

It also reduced the complexity of system-level debugging by allowing individual processing stages to be examined before their integration.

The final development stage therefore focused on **continuous GPS waveform generation, SDR data streaming, RF transmission, and receiver-level validation**.

Chapter 4

Results, Verification, Challenges and Discussion

This chapter presents the experimental results obtained from the developed GPS L1 signal-generation system. The results are presented in the same sequence as the processing architecture described in Chapter 3, beginning with the processing of satellite navigation data and progressing toward PRN generation, carrier generation, navigation-message conversion, integrated GPS waveform generation, SDR transmission, and receiver-level validation.

The purpose of this chapter is not only to show whether each VI executed successfully, but also to explain what each output represents, why the observed result is important, and how it contributes to the complete GPS L1 signal-generation process.

The development was carried out using a modular approach. The RINEX File Parser and GPS Data Sorting VIs established the satellite-data processing stage; the PRN C/A and Carrier Signal Generator VIs were verified independently; and the Navigation Message Conversion/Streaming VI established the transfer of navigation words into LabVIEW. For final end-to-end L1 generation and transmission, the NI GPS Data Streaming VI was used to provide the integrated GPS waveform-generation and USRP interface.

The most significant experimental result was obtained at the receiver. The generated L1 waveform was transmitted through the USRP-2920 and antenna, after which a mobile GPS receiver detected multiple satellites and achieved a 3D position fix.

4.1 Results of RINEX Navigation Data Processing

4.1.1 RINEX File Parser Output

The RINEX File Parser VI represents the first practical stage of the developed signal-generation system. Its purpose is to transform the human-readable RINEX navigation file into numerical satellite information that can be processed by the remaining GPS signal-generation stages.

The parser first separates the RINEX header from the navigation records by locating the `END OF HEADER` marker. The navigation records are then processed and their numerical parameters are extracted for further organization.

The output shown in Figure 4.1 consists of extracted navigation parameters rather than a waveform. This distinction is important because this stage operates entirely in the data-processing domain.

The extracted information represents satellite navigation parameters such as ephemeris and clock-related information. These values are required later because the GPS signal cannot be generated from a PRN code alone; the navigation information provides satellite-specific information that is incorporated into the navigation-message processing chain.

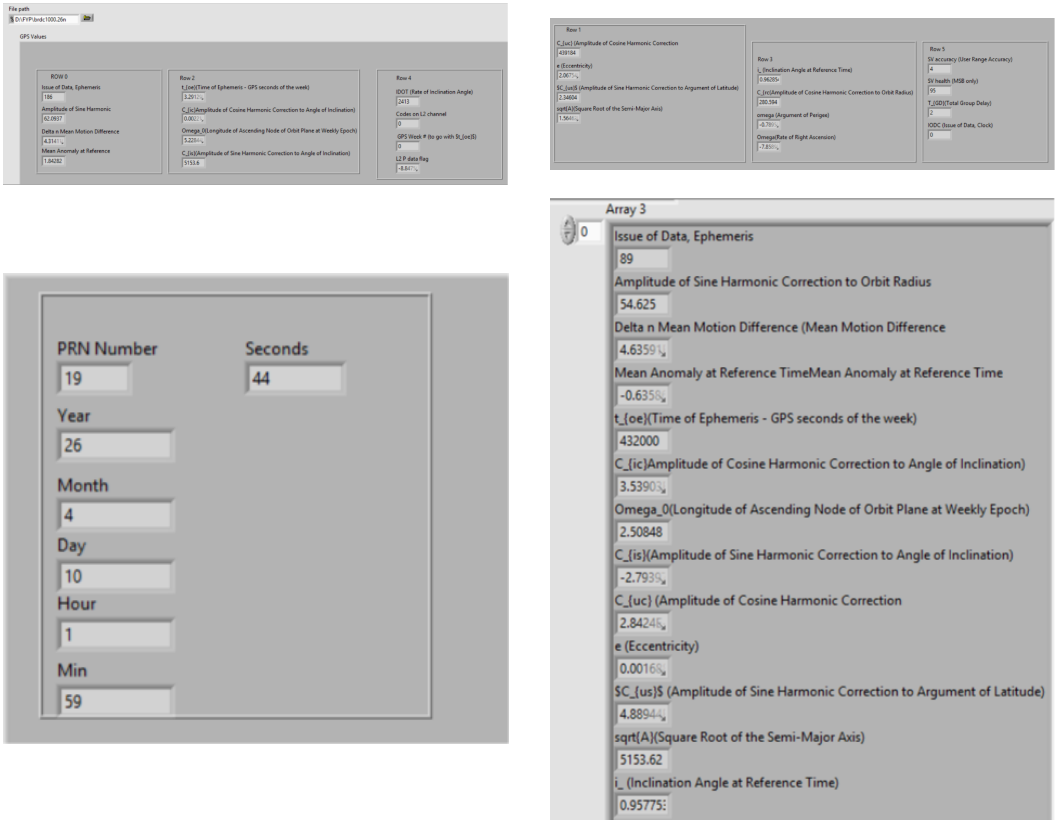


Figure 4.1: Output of the RINEX File Parsing VI showing the extracted satellite navigation parameters.

The output represents the transition from RINEX text data to numerical GPS parameters:

$$\boxed{\begin{array}{c}\text{RINEX Text Data}\\ \downarrow \\ \text{Numerical GPS Parameters}\end{array}} \tag{4.1.1}$$

The successful extraction of these parameters verifies that the RINEX file can be accessed from LabVIEW, its header can be separated from the navigation records, and the numerical satellite parameters can be passed to the subsequent processing stage.

4.1.2 Result Interpretation

The important outcome of this stage is not a single numerical value. Instead, the result demonstrates that:

- the RINEX file can be accessed from LabVIEW;
- the header can be separated from the navigation records;
- navigation records can be parsed;
- numerical satellite parameters can be extracted; and
- the extracted information can be passed to the next processing stage.

The RINEX parser therefore establishes the data source for the GPS signal-generation chain.

4.2 Results of GPS Data Sorting VI

4.2.1 Sorted Satellite Navigation Data

The raw output from the RINEX parser contains the extracted values in the order in which they are obtained from the file. The GPS Data Sorting VI reorganizes these values into structured arrays so that each parameter can be accessed according to its corresponding navigation field.

The thesis implementation identifies the output as organized satellite ephemeris information, which is subsequently made available to the navigation-message processing stage.

The output arrays show that the extracted navigation parameters are no longer treated simply as a continuous sequence of values. Instead, the data are organized into a structured representation.

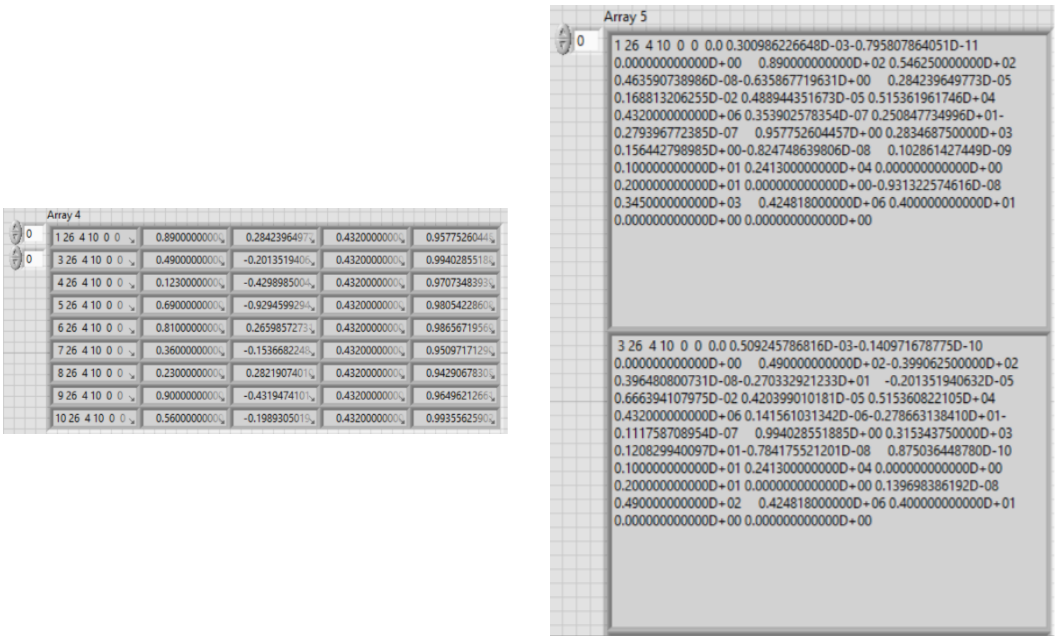


Figure 4.2: Output of the GPS Data Sorting VI showing the organized satellite navigation data.

The project implementation identifies Array 5 as the input ephemeris data extracted from the RINEX file and Array 4 as the sorted GPS ephemeris parameters generated by the GPS Data Sorting VI.

4.2.2 Result Interpretation

This result is important because the Navigation Message stage requires individual satellite parameters in an accessible form.

The processing relationship is:

$$\text{RINEX Parsed Data} \rightarrow \text{Unorganized Parameters} \rightarrow \text{GPS Data Sorting} \rightarrow \text{Structured Satellite Data} \tag{4.2.1}$$

The successful production of the organized arrays demonstrates that the RINEX-processing stage provides a usable interface to the subsequent navigation-message stage.

4.3 Results of PRN C/A Code Generator VI

4.3.1 PRN Code Generation

The PRN C/A Code Generator VI was developed to produce the satellite-specific spreading sequence used by the GPS L1 signal. Each selected GPS satellite is represented by its corresponding PRN sequence, allowing the receiver to distinguish the signal associated with that satellite.

For the L1 C/A signal, the generated sequence contains:

$$N_{\mathrm{C/A}} = 1023 \text{ chips} \tag{4.3.1}$$

and represents one complete C/A code period. The project implementation uses the G1/G2 register concept and PRN-specific selection logic to generate the sequence.

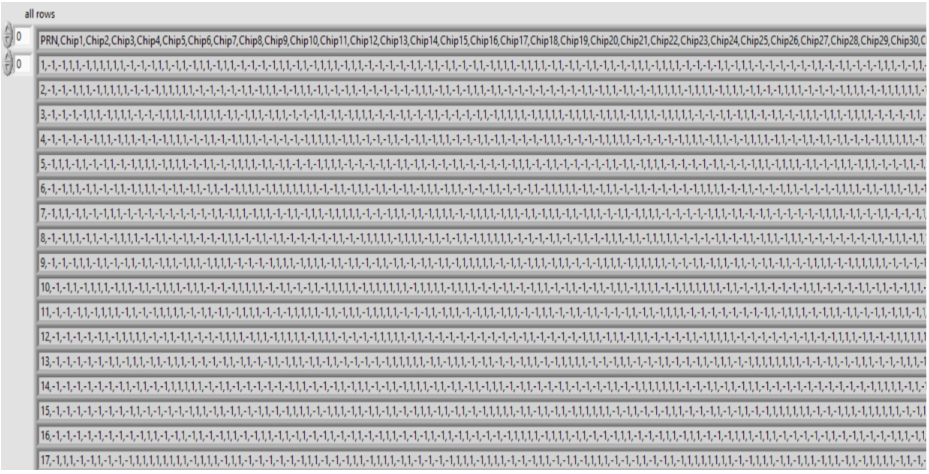


Figure 4.3: Output of the GPS C/A PRN code generation VI showing the generated PRN sequence.

In the displayed output, the chip values are represented using (+1) and (-1). This bipolar representation is useful for subsequent signal-processing operations because it directly provides the two possible signal states used during spreading and BPSK-related processing.

For example, the beginning of a row can be represented conceptually as:

$$[+1,,-1,,-1,,+1,,+1,,-1,\ldots] \tag{4.3.2}$$

The sequence does not appear as a simple repeating pattern. Its pseudo-random appearance is a consequence of the PRN-generation process.

4.3.2 Python Backend and CSV Output

The PRN generation stage also uses a Python implementation as the computational backend. The Python code generates the selected PRN sequence and stores the resulting codes in a CSV file. The CSV representation provides a persistent numerical form of the generated codes that can be inspected independently from the LabVIEW front panel.

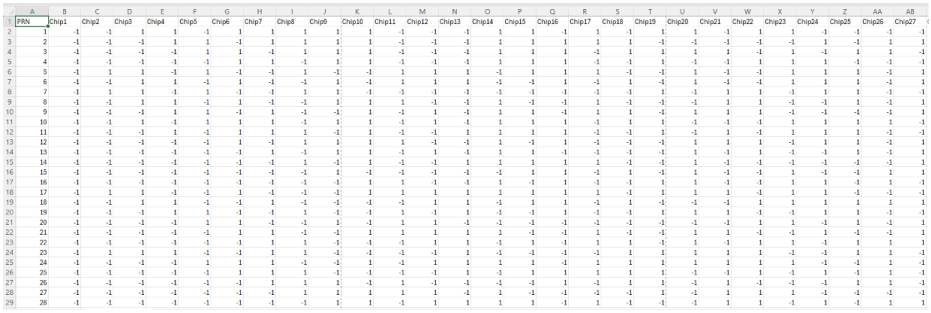


Figure 4.4: CSV representation of the generated GPS C/A PRN codes produced by the Python implementation.

This provides two levels of verification:

1. **Python/CSV verification:** confirms that the numerical PRN sequence has been generated and stored as shown in Figure 4.4..

2. **LabVIEW VI verification:** confirms that the sequence can be accessed and represented within the signal-generation environment.

Thus, the CSV file provides a convenient reference dataset for checking the generated PRN sequences.

4.3.3 Significance of the Output

The result verifies that the developed generator can produce satellite-specific C/A sequences rather than relying on a single hard-coded waveform.

The PRN output therefore provides the spreading-code input required by the subsequent signal-processing chain.

The overall generation process can be represented as:

$$\boxed{\begin{array}{c}\text{Selected PRN} \rightarrow \text{G1/G2 Generation} \\ \rightarrow \text{PRN-Specific Selection} \rightarrow \text{1023-Chip C/A Sequence}\end{array}} \tag{4.3.3}$$

The successful generation and visualization of the sequence confirms the functional operation of the PRN-generation stage.

4.4 Results of Carrier Signal Generator VI

4.4.1 Carrier Waveform Generation

The Carrier Signal Generator VI was developed to independently verify the generation of a sinusoidal carrier waveform before its use in the GPS signal-processing chain.

For module-level verification, the following parameters were used.

Table 4.1: Carrier Signal Generator module-level test parameters.

Parameter	**Test Value**
Carrier frequency	1 MHz
Sampling frequency	20 MHz
Number of samples	10,000
Amplitude	Approximately ± 1
Initial phase	0

The carrier waveform is represented by:

$$x(t) = A\sin\left(2\pi f_c t + \phi\right) \tag{4.4.1}$$

where A is the carrier amplitude, f_c is the carrier frequency, and t is the time variable.

For the module-level test:

$$f_c = 1, \text{MHz} \tag{4.4.2}$$

and

$$f_s = 20, \text{MHz}. \tag{4.4.3}$$

The number of samples per carrier cycle is therefore:

$$N_{\text{samples/cycle}} = \frac{f_s}{f_c} = \frac{20\,\text{MHz}}{1\,\text{MHz}} = 20 \tag{4.4.4}$$

shows a continuous sinusoidal signal oscillating around zero. The vertical axis represents amplitude, while the horizontal axis represents the sample/time representation used by the LabVIEW waveform display.

The waveform alternates between positive and negative values, demonstrating the expected periodic behaviour of the generated carrier.

4.4.2 Verification Against LabVIEW Reference

The custom carrier-generation result was compared with LabVIEW's built-in signal-generation function using equivalent parameters. The comparison provided an independent reference for frequency, amplitude, phase, and waveform behaviour.

The result demonstrates that the carrier-generation algorithm produces the expected sinusoidal waveform and agrees with the LabVIEW signal-generation reference.

It should be noted that the 1 MHz value is a module-level test value.

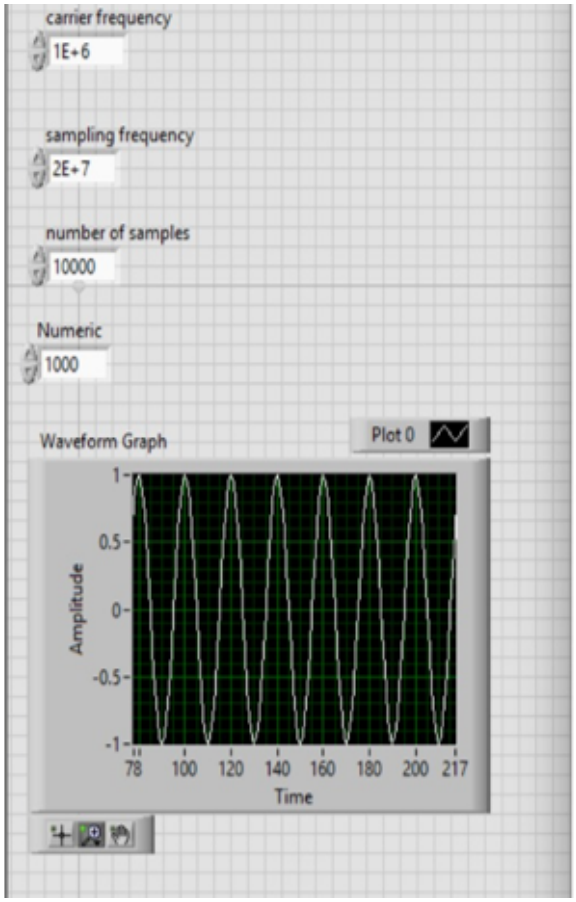


Figure 4.5: Generated GPS L1 carrier signal waveform.

4.5 Results of Navigation Message Conversion and TCP Streaming VI

4.5.1 Navigation Word Reception

The Navigation Message stage was developed further from the initial duration-based bit-generation concept into a TCP-based navigation-word conversion and streaming interface.

The final implementation establishes a TCP connection to the local navigation-message server and receives GPS navigation words as U32 values. These words are then converted into Boolean arrays representing individual navigation bits.

The IP address is configured as:

$$\text{IP} = 127.0.0.1 \tag{4.5.1}$$

and the communication port is:

$$\text{Port} = 9999 \tag{4.5.2}$$

The GPS Word (U32) indicator shows the received navigation word as an unsigned 32-bit numerical value. The Boolean array represents the individual bits extracted from that word.

The conversion process is therefore:

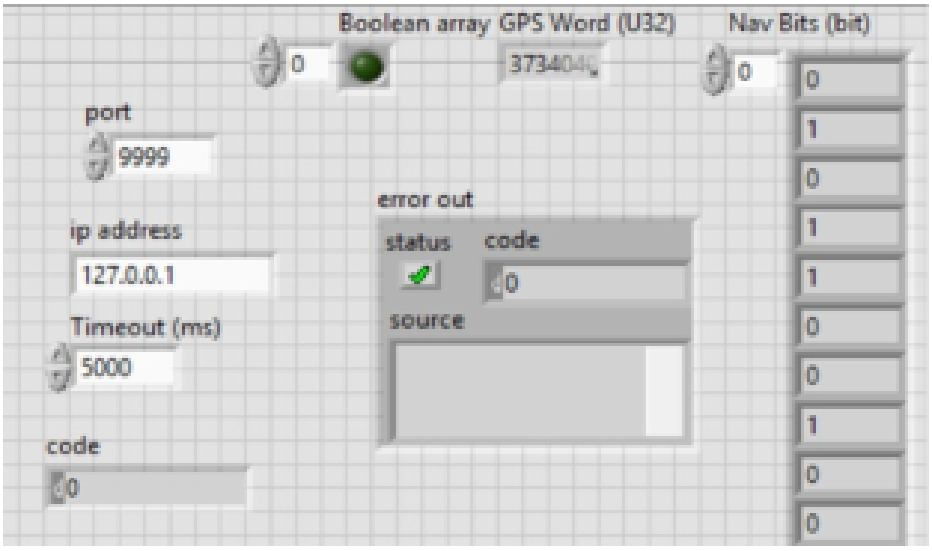


Figure 4.6: Received GPS navigation word displayed as a U32 value in the Navigation Message VI.

$$\boxed{\begin{array}{c}\text{GPS Word (U32)}\\ \downarrow \\ \text{Individual Binary Bits}\end{array}} \tag{4.5.3}$$

4.5.2 Navigation-Bit Waveform

The waveform graph displayed on the front panel represents the resulting navigation-bit sequence. Unlike the carrier waveform, this graph is not expected to appear sinusoidal. It represents a digital sequence, so the signal changes between levels corresponding to binary states.

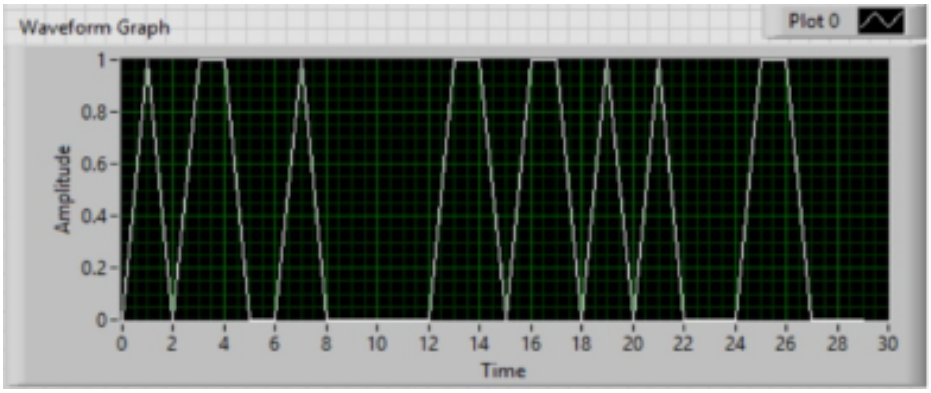


Figure 4.7: Navigation-bit waveform obtained after conversion of the received GPS word into individual binary bits.

The graph therefore provides visual confirmation that the received navigation word has been converted into a bit-oriented representation.

The documented test used the `brdc1000.26n` broadcast-navigation file, PRN 1, GPS week 2413, a time of ephemeris of 432000 s, approximately 0.6 s per word, and a 30-second frame configuration.

4.5.3 Server-Side Result

The corresponding server-side execution is shown in Figure 4.8.

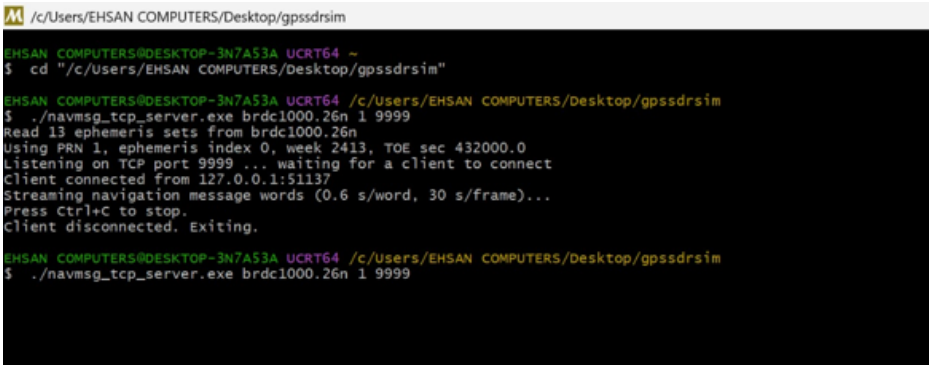


Figure 4.8: Server-side execution showing the generation and TCP streaming of GPS navigation words.

The corresponding server-side execution showed that the navigation-message server:

1. loaded the broadcast ephemeris file;
2. selected the required PRN;
3. established the TCP listening socket;
4. accepted the LabVIEW client connection; and
5. streamed navigation words continuously.

This confirms that the TCP interface operated as an active communication path rather than simply displaying locally generated test data.

4.5.4 Result Interpretation

The output verifies the successful transfer of navigation information from the navigation-data source into LabVIEW.

The complete data path can be summarized as:

$$\boxed{\begin{array}{c}\text{Broadcast Navigation Data} \rightarrow \text{Navigation Server} \\ \rightarrow \text{TCP} \rightarrow \text{GPS Word} \rightarrow \text{Boolean Navigation Bits}\end{array}} \tag{4.5.4}$$

This stage provided the required navigation-bit interface for the integrated GPS signal-generation environment.

4.6 Final GPS L1 Signal Integration Using NI GPS Data Streaming VI

4.6.1 Integrated GPS Generation

After the individual modules were developed, the next stage was to obtain a complete GPS waveform suitable for SDR transmission.

The final integration was performed using the NI GPS Data Streaming VI. The final output stage requires specialized NI blocks protected by National Instruments intellectual property. These blocks cannot be independently copied or recreated as separate user-defined blocks. The NI GPS Data Streaming VI therefore provided the supported integrated path for GPS waveform generation and USRP communication.

The individual project modules remain important because they establish the processing concepts and independently verifiable stages corresponding to the functions contained within the integrated environment.

4.7 NI GPS Data Streaming VI Configuration Results

4.7.1 GPS Scenario Configuration

The NI GPS Data Streaming VI provides a dedicated configuration area for the GPS scenario.

The GPS settings section includes parameters such as:

- almanac file path;
- ephemeris file path;
- number of frames;
- initial GPS time of week;
- optimal satellite update interval;
- maximum number of optimal satellites;
- receiver-location format;
- receiver coordinates;
- reference power level;
- maximum relative power level; and
- waveform scaling factor.

These settings determine the simulated satellite scenario from which the GPS waveform is generated.

The ephemeris and almanac files provide satellite information required by the integrated GPS generator. The initial GPS time establishes the simulated time reference, while the number of frames determines the amount of generated navigation/signal data.

The receiver location is particularly important because it defines the position represented by the generated GPS scenario. This configured location is later compared with the position displayed by the receiver.

4.7.2 Interpretation of GPS Scenario Configuration

The GPS configuration panel therefore establishes the satellite scenario represented by the generated signal. The major relationship can be summarized as:

$$\boxed{\begin{array}{c}\text{GPS Data} + \text{Time Parameters} + \text{Receiver Scenario} \\ \rightarrow \text{Configured GPS Environment}\end{array}} \tag{4.7.1}$$

4.8 USRP Transmission Configuration

- device identification or IP address;
- carrier frequency;
- IQ sample rate;
- gain;
- frequency offset;
- reference-frequency source;
- satellite selection;
- generation state; and
- producer/consumer queue status.

an IQ rate of approximately $4\,\mathrm{MS/s}$, and a gain of $30\,\mathrm{dB}$. The selected satellites are also displayed in the panel. This confirms that the integrated generator is configured according to a defined satellite scenario.

The relationship between the scenario and transmission configuration can be represented as:

$$\boxed{\begin{array}{c}\text{GPS Scenario} + \text{Signal Parameters} + \text{USRP Configuration} \\ \rightarrow \text{Transmit-Ready GPS Waveform}\end{array}} \tag{4.8.1}$$

4.9 Producer–Consumer Architecture and Streaming Result

The NI GPS Data Streaming VI uses a producer–consumer architecture to separate waveform generation from waveform transmission.

4.9.1 Producer Loop

The Producer Loop is responsible for generating GPS waveform data in manageable blocks.

The process can be represented as:

$$\text{GPS Scenario} \rightarrow \text{Generate GPS Waveform Block} \rightarrow \text{Place Block in Queue} \rightarrow \text{Continue Generation}. \tag{4.9.1}$$

The queue provides temporary storage between waveform generation and transmission.

4.9.2 Consumer Loop

The Consumer Loop retrieves waveform blocks from the queue and transfers them toward the USRP.

The process is:

$$\text{Queue} \rightarrow \text{Retrieve Waveform Block} \rightarrow \text{Display/Monitor} \rightarrow \text{Send Data to USRP}. \tag{4.9.2}$$

This architecture allows waveform generation and transmission to operate as separate processes. The Producer does not have to wait for the complete transmission of one block before generating the next one.

4.9.3 Result Interpretation

The producer–consumer architecture is important because GPS transmission requires a continuous data stream. Generating and transmitting the signal in independent blocks makes the data flow more manageable and reduces the possibility that processing delays in one part of the system will interrupt the other.

When the system is stopped, the remaining queued data are processed and the queue and associated GPS/USRP sessions are closed in an orderly manner.

4.10 Final Generated GPS L1 Waveform

4.10.1 Time-Domain Waveform

The carrier-generator result represents only a sinusoidal carrier. In contrast, the waveform displayed by the NI GPS environment represents the generated GPS baseband/IQ samples after GPS navigation-data processing, PRN spreading, and modulation. Therefore, the graph should not be interpreted by reading its horizontal axis as 1575.42 MHz. Instead, it represents sampled waveform data in the time domain.

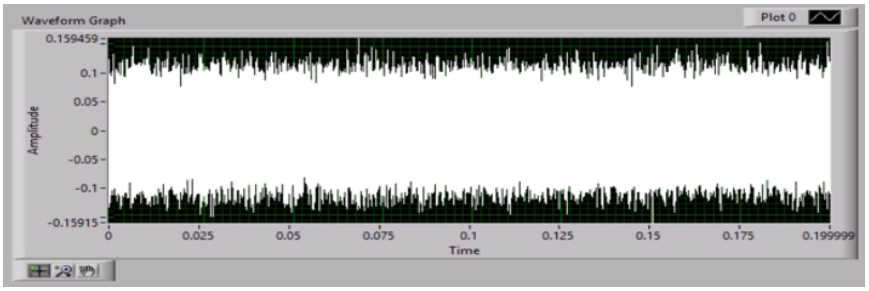


Figure 4.9: Generated GPS signal waveform in the time domain.

The rapidly changing waveform is a consequence of the digital GPS signal structure, including navigation-data variation, PRN code transitions, spreading, modulation, and sampled I/Q waveform generation.

4.10.2 Interpretation of the Waveform Shape

The waveform does not appear as a smooth sinusoid because it represents the processed GPS signal samples rather than the carrier alone.

The distinction between the two outputs is therefore:

$$\text{Standalone Carrier} \rightarrow \text{Sinusoidal Reference} \tag{4.10.1}$$

whereas:

$$\text{Final GPS Waveform} \rightarrow \text{Modulated and Spread Signal} \tag{4.10.2}$$

The final waveform can be considered as the combination of:

$$\text{Navigation Data} + \text{PRN Spreading} + \text{Modulation} + \text{Carrier Representation} \tag{4.10.3}$$

The waveform result therefore confirms the generation of a complex GPS signal representation before RF transmission.

4.11 Zoomed View of the Generated Waveform

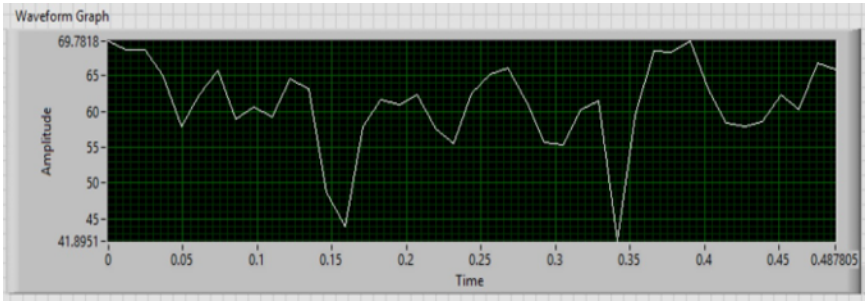


Figure 4.10: Zoomed View of Generated GPS signal waveform in the time domain.

The zoomed graph provides a clearer view of the rapid sample-to-sample variations present in the generated GPS waveform.

At the larger scale, the waveform contains positive and negative excursions and rapid transitions. These transitions are a consequence of the underlying digital signal processing and should not be interpreted as noise or an unstable carrier by themselves.

The zoomed view demonstrates that the waveform consists of continuously varying sampled signal data rather than a constant DC value or an unmodulated sinusoid.

4.12 USRP-2920 Transmission Result

The generated GPS waveform was transferred from the LabVIEW environment to the USRP-2920 through the configured NI-USRP interface.

The complete transmission path is:

$$\boxed{\begin{array}{c}\text{LabVIEW} \rightarrow \text{NI GPS Data Streaming VI} \rightarrow \text{NI-USRP Interface} \\ \rightarrow \text{USRP-2920} \rightarrow \text{TX Antenna} \\ \rightarrow \text{GPS L1 Signal} \rightarrow \text{GPS Receiver}\end{array}} \tag{4.12.1}$$

The USRP-2920 served as the SDR transmitter and provided the interface between the generated digital waveform and the RF transmission chain.

The importance of this result is that the project progressed beyond software waveform visualization. The generated signal was transferred to SDR hardware and presented to the RF transmission chain.

4.13 Spectrum Analyzer Verification

After transmission through the USRP-2920, the RF output was observed using a spectrum analyzer to verify the presence of the generated GPS L1 signal in the frequency domain. This measurement provides an intermediate validation between the digitally generated waveform and the receiver-level results.

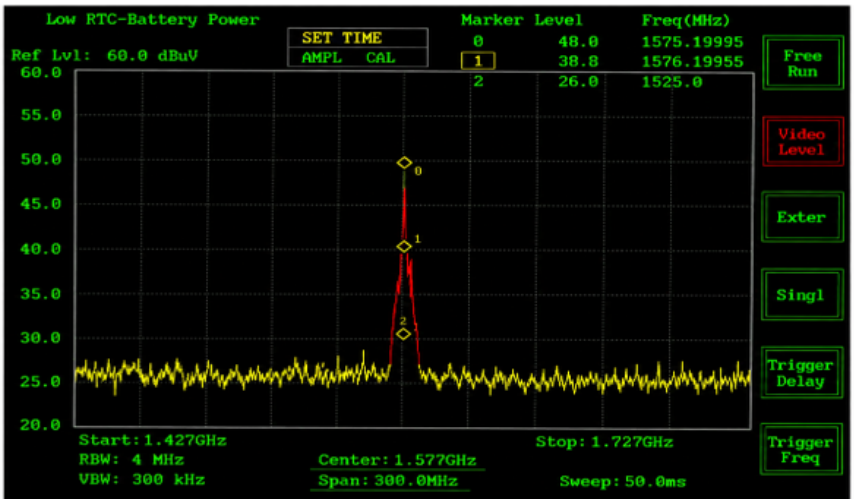


Figure 4.11: Spectrum analyzer measurement showing the transmitted GPS L1 signal.

The spectrum analyzer measurement shows a distinct spectral peak around the GPS L1 center frequency of 1575.42 MHz. The observed peak confirms that the generated signal was successfully transferred through the USRP-2920 and appeared at the expected RF frequency.

The spectrum measurement is different from the time-domain waveform shown in Figure 4.9. The time-domain result verifies the generated sampled waveform, whereas the spectrum analyzer verifies its frequency-domain location after RF transmission.

4.14 Receiver-Level Validation

4.14.1 Satellite Detection

The mobile receiver was used to determine whether the transmitted signal could be acquired and processed by a real GPS receiver.

This observation is significant because it demonstrates that the receiver was able to detect GPS signal information and perform the acquisition and tracking processes required before calculating a position.

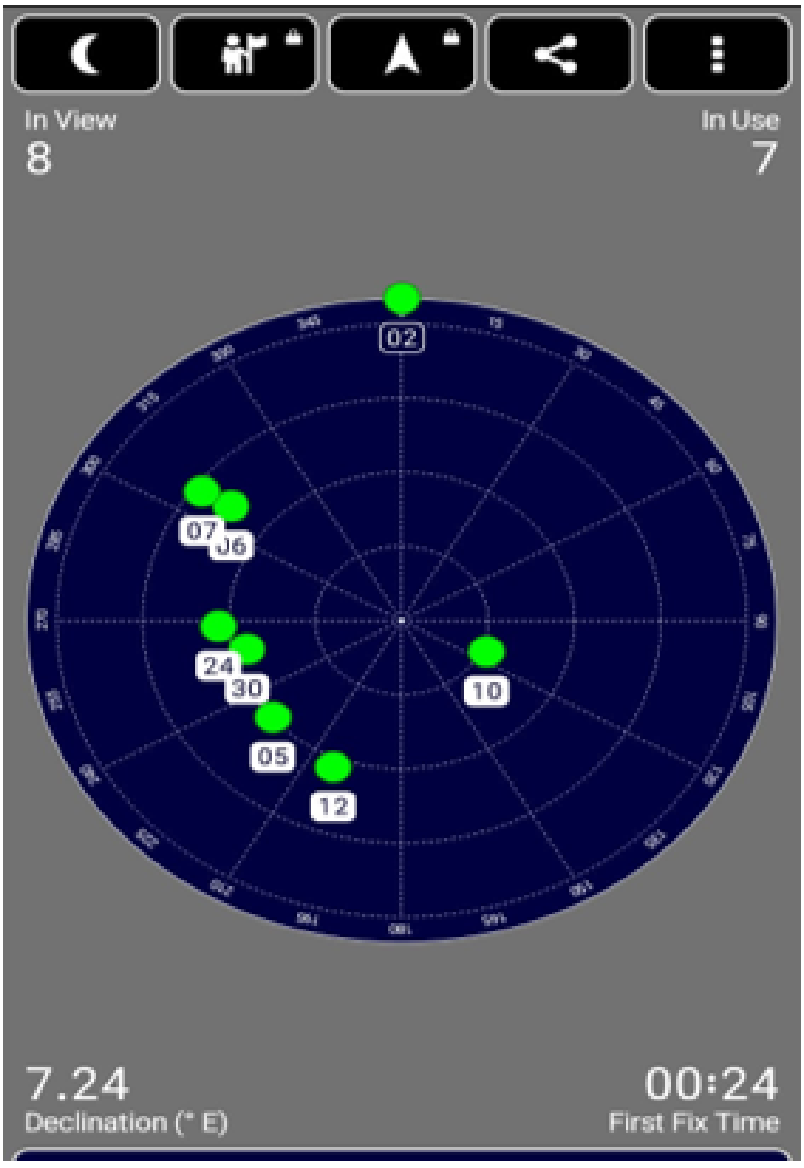


Figure 4.12: Satellite detection at the GPS receiver level.

The final experimental result therefore provides receiver-level evidence rather than only waveform-level evidence.

4.14.2 Position Fix Result

The second receiver result is the displayed position solution.

The GPS Test application reports a **3D Fix**, indicating that the receiver was able to use the acquired satellite measurements to calculate a three-dimensional position solution.

The application also displays the resulting latitude and longitude on the map interface.



Figure 4.13: Satellite detection at the GPS receiver level.

The experimental scenario was configured in LabVIEW with a specified receiver location. The final experimental result indicates that the position displayed by the receiver corresponded to the coordinates configured in the GPS scenario.

4.14.3 Significance of the 3D Fix

The 3D fix is the strongest receiver-level result obtained during the project.

The validation chain can be expressed as:

$$\boxed{\begin{array}{c}\text{GPS Scenario} \rightarrow \text{GPS Waveform} \rightarrow \text{USRP-2920} \\ \downarrow \\ \text{RF Transmission} \rightarrow \text{Receiver Acquisition} \\ \downarrow \\ \text{Satellite Detection} \rightarrow \text{3D Position Fix}\end{array}} \tag{4.14.1}$$

Each stage represents a further level of system validation.

The receiver result demonstrates that the transmitted L1 signal was sufficiently structured to be recognized and processed by a GPS receiver.

4.15 Overall Experimental Verification

The results obtained from the complete GPS signal generation and transmission system are summarized in Table 4.2. The verification covers the signal-processing stages, digital waveform generation, RF transmission, frequency-domain measurement, and receiver-level validation.

The experimental results demonstrate a progressive validation of the developed system. The RINEX parser and GPS data-processing modules provided the required navigation information, while the PRN, carrier, navigation-data, and BPSK modulation stages generated the individual components required for the GPS L1 signal. These components were subsequently integrated through the NI GPS Data Streaming VI to produce the final GPS waveform.

The generated waveform was transferred to the USRP-2920 for RF transmission. The spectrum analyzer measurement provided frequency-domain verification by showing a distinct peak around the GPS L1 center frequency of 1575.42 MHz. Finally, the mobile GPS receiver detected multiple satellites and obtained a 3D position fix, providing receiver-level evidence that the transmitted signal could be acquired and processed.

4.16 Challenges Encountered and Their Resolution

4.16.1 RINEX Data Handling

RINEX navigation files are text-based and contain a header followed by structured navigation records. Directly using these records in the signal-generation process would make the subsequent LabVIEW implementation unnecessarily complicated.

The solution was to separate the processing into the RINEX File Parser VI and GPS Data Sorting VI. This created a clear boundary between file interpretation and GPS signal processing.

4.16.2 Navigation Message Processing

The navigation message required a controlled method for transferring GPS words into LabVIEW.

The TCP-based approach provided a practical interface between the navigation-message source and LabVIEW. GPS words could be received as numerical values and converted into Boolean arrays for visualization and subsequent processing.

Table 4.2: Summary of experimental results and their significance.

Processing Stage	Observed Result	Significance
RINEX Parser	Navigation parameters successfully extracted	Provides the required satellite navigation data.
GPS Data Sorting	Navigation parameters organized into structured arrays	Provides correctly formatted data for subsequent processing.
PRN Generator	1023-chip bipolar PRN sequence generated	Confirms generation of the satellite-specific spreading code.
Carrier Generator	Periodic L1 carrier waveform generated	Verifies the carrier-generation stage at the GPS L1 frequency.
Navigation Data	GPS navigation words converted into a digital bit stream	Provides the required navigation information for signal modulation.
BPSK Modulation	Navigation data and PRN code combined through modulation	Implements the required GPS spreading and modulation process.
NI GPS Data Streaming	Integrated GPS L1 baseband/IQ waveform generated	Confirms successful integration of the GPS signal-generation stages.
Waveform Display	Rapidly varying time-domain GPS waveform observed	Confirms generation of the processed GPS signal before RF transmission.
USRP-2920	Generated waveform transferred to the SDR	Establishes the digital-to-RF transmission path.
Spectrum Analyzer	Distinct spectral peak observed around 1575.42 MHz	Confirms the presence of the transmitted signal in the GPS L1 band.
GPS Receiver	Multiple GPS satellites detected	Demonstrates successful receiver-level signal acquisition.
GPS Receiver	3D position fix obtained	Provides final receiver-level validation of the transmitted GPS signal.

This approach also made the data stream observable during debugging.

4.16.3 Integration of Individually Developed VIs

A major engineering challenge was integrating individually developed GPS-processing blocks into a complete transmitter.

The project addressed this through the NI GPS Data Streaming VI. Specialized NI blocks required for the final output stage are protected by National Instruments intellectual property and therefore cannot be copied or independently recreated. The integrated NI VI provides the supported path for final signal generation and transmission.

This allowed the project to retain the independently developed VIs as the understanding, development, and verification foundation, while using the integrated NI environment for the final end-to-end L1 experiment.

4.16.4 Continuous Data Streaming

Generating a GPS waveform and transmitting it continuously are different requirements. A waveform may be correctly generated but still fail to transmit continuously if the data-processing and hardware-transfer stages are not synchronized.

The producer–consumer architecture addresses this problem by separating waveform generation from waveform transmission through a queue.

This architecture provided the continuous data flow required for the final SDR experiment.

4.17 Discussion of Experimental Results

The results demonstrate a progression from individual algorithmic outputs to complete receiver-level validation.

The first stages—RINEX parsing, GPS data sorting, and PRN generation—operate primarily on numerical data. Their outputs are therefore arrays and structured parameters rather than RF waveforms. These stages establish the information required to construct the GPS signal.

The carrier-generation stage demonstrates the transition from numerical parameters to a sampled periodic signal. Its agreement with the LabVIEW reference confirms the correctness of the basic waveform-generation approach.

The Navigation Message Conversion/Streaming stage introduces time-varying navigation information into the processing environment. Its output demonstrates that GPS words can be received and converted into individual binary states.

The final NI GPS waveform is fundamentally different from each individual output. It represents the combined GPS signal-processing result and is therefore expected to appear considerably more complex than either the PRN sequence or the standalone carrier.

The final waveform confirms that a continuously varying GPS signal representation is generated before transmission. The waveform is interpreted as a baseband/IQ representation rather than a direct visualization of the 1575.42 MHz RF carrier.

The most important result is the receiver response. The mobile receiver detected multiple satellites and obtained a 3D fix, with the displayed position corresponding to the coordinates configured in the LabVIEW scenario.

The experimental validation can therefore be considered at three levels:

1. **Level 1 — Algorithm/Module Verification:** The individual VIs produced their intended intermediate outputs.

2. **Level 2 — Integrated Waveform Verification:** The NI GPS environment produced a continuous GPS waveform suitable for transmission.

3. **Level 3 — Receiver-Level Verification:** The transmitted signal was detected by a GPS receiver and resulted in a 3D position solution.

The third level is particularly important because it demonstrates practical operation of the complete L1 transmission chain rather than only the internal software behaviour.

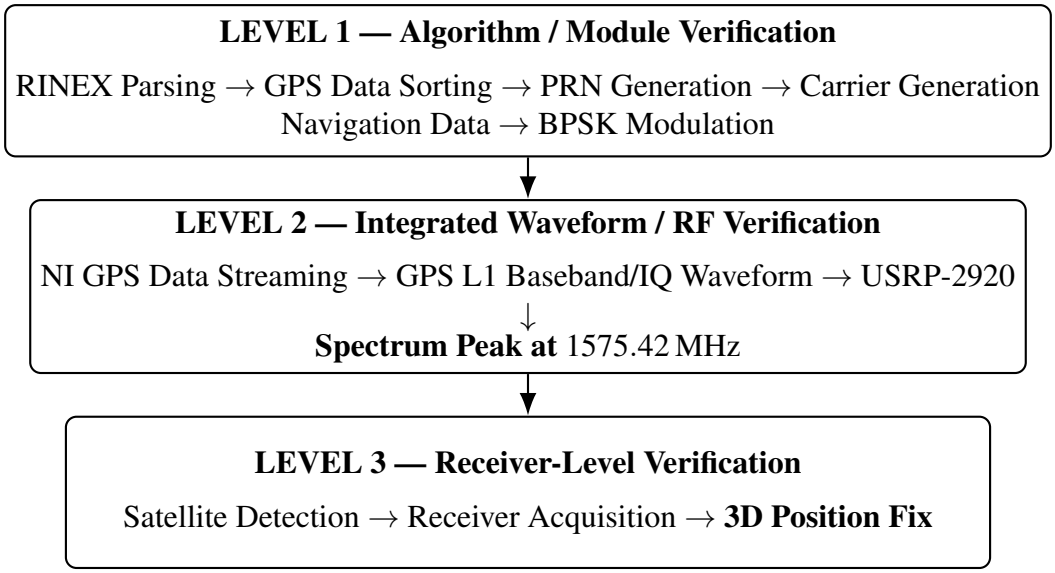


Figure 4.14: Three-level experimental validation of the developed GPS L1 signal generation and transmission system.

4.18 Limitations of the Experimental Results

The results should be interpreted within the actual scope of the completed implementation.

The experimentally validated system is a GPS L1 implementation. Although the overall project objectives include L1, L2, and L5, L2 and L5 were not generated and experimentally validated during the reported project phase. They should therefore not be presented as achieved experimental results.

Similarly, the receiver experiment demonstrates successful acquisition and position fixing, but it should not be interpreted as a complete quantitative characterization of GPS receiver performance. Measurements such as detailed C/N_0 analysis, repeated statistical accuracy measurements, spectrum characterization, and comparative L1/L2/L5 performance remain suitable areas for future work.

The receiver result should therefore be described precisely as:

> Successful laboratory validation of the GPS L1 signal-generation and transmission chain through receiver satellite detection and 3D position fixing.

Chapter 5

Conclusion

The experimental work presented in this chapter demonstrates the progressive validation of the developed GPS L1 signal-generation system.

The RINEX File Parser successfully converted navigation-file information into numerical satellite parameters, and the GPS Data Sorting VI organized these parameters into a form suitable for subsequent GPS processing. The PRN C/A Code Generator produced the required satellite-specific 1023-chip bipolar sequence, while the Carrier Signal Generator produced and independently verified the sinusoidal carrier waveform.

The Navigation Message Conversion/Streaming VI established a TCP-based interface through which GPS navigation words were received and converted into individual navigation bits. These independently developed modules provided the fundamental processing and verification framework for the GPS L1 signal chain.

For final system integration, the NI GPS Data Streaming VI provided the integrated GPS waveform-generation and streaming functionality required for the SDR implementation. Its producer–consumer architecture generated waveform blocks, temporarily stored them in a queue, and transferred them to the USRP transmission path.

The resulting GPS waveform was successfully observed in LabVIEW. The waveform represented the generated GPS baseband/IQ samples following navigation-data processing, PRN spreading, and modulation, rather than a direct visualization of the L1 RF carrier.

The generated waveform was then transmitted through the USRP-2920 and antenna. At the receiver, multiple satellites were detected and a 3D position fix was obtained. The receiver displayed the position corresponding to the configured GPS scenario, providing the final receiver-level validation of the implemented L1 transmission chain.

Therefore, the principal experimental achievement of the project is the development and laboratory validation of a GPS L1 signal-generation and transmission system using LabVIEW, NI GPS Data Streaming functionality, USRP-2920 SDR hardware, and a mobile GPS receiver. L2 and L5 remain future extensions of the developed architecture rather than completed experimental components.

Appendix A

Python Backend Implementation of the PRN C/A Code Generator

The PRN C/A Code Generator VI uses a Python-based backend for generating satellite-specific GPS C/A codes. The backend generates the required PRN sequence based on the selected satellite PRN number and provides the generated code for integration with the LabVIEW implementation.

The following figures present the Python code used in the backend implementation.

```python
import numpy as np
import csv


# GPS PRN tap pairs
TAPS = {
    1:(2,6),  2:(3,7),  3:(4,8),  4:(5,9),  5:(1,9),
    6:(2,10), 7:(1,8),  8:(2,9),  9:(3,10), 10:(2,3),
    11:(3,4), 12:(5,6), 13:(6,7), 14:(7,8), 15:(8,9),
    16:(9,10),17:(1,4), 18:(2,5), 19:(3,6), 20:(4,7),
    21:(5,8), 22:(6,9), 23:(1,3), 24:(4,6), 25:(5,7),
    26:(6,8), 27:(7,9), 28:(8,10),29:(1,6), 30:(2,7),
    31:(3,8), 32:(4,9)
}


def generate_ca(prn):

    if prn not in TAPS:
        raise ValueError("PRN must be between 1 and 37")

    g1 = np.ones(10, dtype=int)
    g2 = np.ones(10, dtype=int)

    ca = []

    tap1, tap2 = TAPS[prn]
```

```python
    for _ in range(1023):

        g1_out = g1[9]
        g2_out = g2[tap1 - 1] ^ g2[tap2 - 1]

        ca_bit = g1_out ^ g2_out

        # Convert GPS chip:
        # 0 -> +1
        # 1 -> -1
        bpsk_chip = 1 if ca_bit == 0 else -1

        ca.append(bpsk_chip)


        g1_feedback = g1[2] ^ g1[9]

        g2_feedback = (
            g2[1] ^
            g2[2] ^
            g2[5] ^
            g2[7] ^
            g2[8] ^
            g2[9]
        )
```

```python
        g1[1:] = g1[:-1]
        g2[1:] = g2[:-1]

        g1[0] = g1_feedback
        g2[0] = g2_feedback


    return ca


def generate_all_to_csv(filepath):

    with open(filepath, "w", newline="") as f:

        writer = csv.writer(f)


        # Header row
        writer.writerow(["PRN"] + [f"Chip{i}" for i in range(1, 1024)])


        for prn in range(1, 38):

            code = generate_ca(prn)

            row = [prn]
```

```python
            # Now writing +1 / -1 values
            for chip in code:
                row.append(chip)


            writer.writerow(row)



def main():

    output_file = r"C:\Users\Waram Najam\Desktop\PRN-Project\prn_output.csv"

    generate_all_to_csv(output_file)

    print("DONE - File saved:")
    print(output_file)


if __name__ == "__main__":
    main()
```

Figure A.1: Python backend code used for the PRN C/A Code Generator VI.

References

[1] T. Pany, D. Akos, J. Arribas, M. Z. H. Bhuiyan, P. Closas, F. Dovis, I. Fernandez-Hernandez, C. Fernández-Prades, S. Gunawardena, T. Humphreys, Z. M. Kassas, J. A. L. Salcedo, M. Nicola, M. L. Psiaki, A. Rügamer, Y.-J. Song, and J.-H. Won, "Gnss software-defined radio: History, current developments, and standardization efforts," *NAVIGATION: Journal of the Institute of Navigation*, vol. 71, no. 1, pp. 1–45, 2024.

[2] M. Majoral, C. Fernández-Prades, and J. Arribas, "A flexible system-on-chip field-programmable gate array architecture for prototyping experimental global navigation satellite system receivers," *Sensors*, vol. 23, no. 23, p. 9483, 2023.

[3] D. Forero, S. Esteban, and Ó. Rodríguez-Polo, "Framework to emulate spacecraft orbital positioning using gnss hardware in the loop," *Sensors*, vol. 23, no. 2, p. 885, 2023.

[4] Ibrahem, Hendy, Ali, and Kamel, "Software-defined radio-based simulation and processing of gps l1 signals for leo satellites navigation," in *2025 15th International Conference on Electrical Engineering (ICEENG)*, 2025, pp. 500–505.

[5] J. Kim, S. Lee, B. G. Kim, H. S. Seo, and J. Ahn, "Design and implementation of a gnss receiver development platform for multi-band signal processing," *Journal of Positioning, Navigation, and Timing*, vol. 13, no. 2, pp. 149–158, 2024.

[6] S. Kim, N. Gault, Y. Jo, H. Yoon, B. Park, A. Garcia-Pena, C. Macabiau, and D. M. Akos, "Gnss l5/e5a code properties in the presence of a blanker," *NAVIGATION: Journal of the Institute of Navigation*, vol. 72, no. 2, pp. 1–15, 2025.

[7] S. Kawamoto, N. Takamatsu, and S. Abe, "Ringo: A rinex pre-processing software for multi-gnss data," *Earth, Planets and Space*, vol. 75, no. 54, 2023.

[8] K. Stankevičius and R. Aleksiejūnas, "Low-cost sdr-based gnss signal recorder and replayer for equipment testing," *The Journal of Navigation*, vol. 78, no. 2–3, pp. 339–357, 2025.